

Customer Contact Operating Environment User Interface Specification—Sales Helper

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Prepared by

U S WEST Technologies

Abstract

This document describes the user interface for CCOE Mass Market Consultative Sales Helper for Small Business Group (SBG) and Home and Personal Services (H&PS). The system is designed to meet the user requirements of CCOE dated March 1, 1994. It will have to be modified to take into account new Future Mode of Operation (FMO) requirements. The document also describes how Sales and Service Consultants (SSC) use input devices such as keyboards, a mouse, and screens.

Version Notice

All revisions made to this document are listed below in chronological order. The release number of this document does not relate to the release number of the application:

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1. Introduction

This document describes and defines a graphical user interface (GUI) that supports and facilitates consultative sales of Sales and Service Consultants (SSCs) using knowledge-based technology.

1.1 Purpose

The purpose of this document is to translate the Mass Market Customer Contact Operating Environment (CCOE) Consultative Sales requirements, as defined in the CCOE user requirements document, into user interface specifications.

The design and development of the sales helper for Small Business Group (SBG) can be dated back to Sales and Service Negotiation and Quality (SSNQ). Several prototypes and usability studies have been developed and conducted. Similar stand-alone systems have also been designed, implemented and deployed in the Home Office Channel of SBG and the Priority Channel of H&PS. Usability studies have also been conducted for those systems. For clarity and historical reasons, the Sales Helper for SBG will be used interchangeably with Target Information and Profile System (TIPS), and the Consultative Sales Profile Environment (CSPE) will be used in the same way as the Sales Helper for H&PS.

The purpose of the CCOE Sales Helper is to collect customer site profile items for mass market customers and then to generate solution recommendations based on the needs and profile of the customer at that site. The interface provides a user layer for SSCs to initiate, conduct, and facilitate consultative sales activities. Although the CCOE Sales Helper application focuses on support for Mass Market requirements, other market units may find that this user interface design extends to their requirements.

This document describes the specifications for the user interface. It does not address the CCOE Consultative Sales functionality or design. The *Customer Contact Operating Environment Consultative Sales Functional Specification* provides detailed information on the functional specifications for the CCOE Consultative Sales application. This particular version is an interim draft release of the Sales Helper. The design of CCOE Sales Helper will have to be modified to accommodate FMO (Future Mode of Operations) requirements.

1.2 Authors

The author of this document is Yat-Sang Hung, of the Technical Staff, Shared Applications and Human Factors Engineering, U S WEST Technologies. The author is grateful for the input, comments, and feedback from the CCOE team members and from co-workers in the Knowledge Based Engineering Group and the Human Factors Engineering Group of U S WEST Technologies.

1.3 Audience

This document is intended for use by Mass Market personnel, CCOE project management, user interface designers, functional analysts, developers, testers, and trainers.

1.4 Content

Chapter 2., Scope describes the CCOE Sales Helper functions and user interface. Chapter 3., Usage Assumptions describes the current systems used by SSCs and presents the assumptions about users, work environment, and hardware environment. Chapter 4., Design Conventions discusses the CCOE design philosophy and presents the standard screen and window components and user inputs that support the philosophy. Chapter 5., Business Functions and Processes contains a more detailed description of the supported tasks and the functional organization of the user interface, and Chapter 6., Sales Helper Interface Specification contains the detailed interface specifications of the profile information windows. The Recommendation section in chapter six contains the interface specifications for the Recommendation window and its link to the Product Manager. Chapter 7., Help , Chapter 8., Error Conditions , and Chapter 9., Usability Criteria cover those areas in some detail. Chapter 10., Performance and Quality Criteria and Chapter 11., Field Evaluation Plan will be completed for future releases of this document. Chapter 12., Issues identifies currently unresolved issues for the Sales Manager. Appendix A. contains the H&PS category and item descriptions for the profile windows, and Appendix B. contains the same information SBG.

1.5 Project Description

The Sales Helper (TIPS/CSPE) system is part of the overall CCOE system identified through the re-engineering effort and is sponsored by both SBG and H&PS of U S WEST Communications Inc. It forms one of the basic components in attaining total customer satisfaction by capturing, distributing, and utilizing expert consultative sales knowledge through the development of a knowledge-based system.

The overall objective of the Sales Helper is to organize, represent and present sales, product, and marketing knowledge in a form that can be shared, collected, communicated, and utilized effectively by SSCs and other departments involved.

In the initial design and implementation, the system will be an intelligent sales tool that is fully and seamlessly integrated with the Customer Contact Operating Environment (CCOE). It will generate customized product recommendations based on the telecommunication needs of the customers.

Because of the possible changes of scope of CCOE, the Sales Helper may have to be redesigned to take into account the FMO requirements identified in the business re-engineering process. This interim release of the user interface specification serves both as a closure for the requirements of the transition from the Present Mode of Operation (PMO+) to the FMO and as a spring board to the FMO.

The Sales Helper user interface will be developed using X-Windows technology which conforms to OSF/Motif standards. In addition to the keyboard, SSCs will use a mouse as a pointing device to control the display of information on their windows.

2. Scope

This section describes the CCOE Sales Helper functions and user interface.

2.1 Sales Helper

The Sales Helper of CCOE enables the SSC to gather pertinent profile data about the customer site and to have that data stored on the appropriate customer Corporate Subject Data Base (CSDB). The CCOE Sales Helper application allows the SSC to collect, view and modify profile information about customer sites in the Mass Market customer CSDB for new and existing customers. This includes customer site layout, different customer needs and concerns, customer equipment, and marketing information that has not previously been collected and retained in mass markets. Additionally, the system will generate a set of possible recommended solutions to fulfill the needs of the customers from the profile information the SSC collects.

The Sales Helper of CCOE incorporates a knowledge-based sales assistant system jointly developed and reused for different market channels, such as the Consultative and Sales Helper (CASH) for the Home Office Channel of SBG and the Priority Channel of H&PS. The latter two have been developed and deployed. Specifically, CASH has undergone more than two iterations since its deployment in March, 1994. Usability studies have been conducted to identify areas of improvement and to help the design of the Sales Helper in CCOE.

The CCOE Sales Helper application supports the following functions:

- collecting, displaying, and updating customer site profile items
The profile items are organized in natural and logical categories to aid navigation.
- displaying summary customer site information
- generating and displaying a set of solution recommendations organized around specific capabilities that the solution offers
The generated solutions will not display unavailable products, and the customer's existing products at that site will be tagged for identification.
- displaying the need trace of specific capability as related to the profile items
- displaying benefits of the recommended solutions
- invoking the CCOE product manager from the set of products proposed by the SSCs or directly from the recommendation window.

See the *Customer Contact Operating Environment Sales Helper Functional Specification* for the specification of these major business functions.

2.2 User Interface

When fully implemented, the CCOE application will provide a GUI for consultative sales and service order negotiation. It will provide a presentation layer for SSCs to perform consultative sales and negotiate service while online with customers. CCOE will utilize CSDBs and existing host systems to provide SSCs with functionality that better matches the current job functions. CCOE will replace the current Service Order Negotiation And Retrieval system (SONAR).

The CCOE Sales Helper application provides a sophisticated user interface engineered to assist Mass Market SSCs in their sales and service negotiation tasks. It provides screens specifically designed for information display and for the data entry associated with each consultative task. It also allows SSCs to view multiple windows within CCOE simultaneously and to move easily between different screens. This interaction allows SSCs to establish rapport with the customers, to gather pertinent profile information, to recommend a set of possible solutions fulfilling the needs of the customers, and prepare service orders while online with customers.

The CCOE Sales Helper application implements a GUI that provides multiple windows of information engineered specifically to assist in consultative sales. Each window is modularly developed as a stand-alone piece that supports a specific task or function and that works independently of the entire interface.

SSCs control the display of windows through mouse pointing devices and keyboard input. SSCs can move quickly between windows, depending on how the customer directs the contact. SSCs perform much of the data entry by selecting items in menus that contain only valid options. Popup error dialog boxes provide immediate feedback concerning incorrect or anomalous data. Online help assists new SSCs in using the interface.

3. Usage Assumptions

This chapter describes the systems the SSCs currently use. It then details the assumptions about the users, their expected usage of CCOE Sales Helper, and their work environment. It also includes assumptions about the hardware environment and the applications that SSCs concurrently access from that platform.

3.1 Existing Systems

Currently, two sales helper systems have been deployed in the Home Office Channel of SBG (Phoenix) and the Market Place Channel of H&PS (Omaha). They are stand-alone systems without any interface to legacy systems.

In service centers without a sales helper system, the SSCs have to write down customer profile information on pieces of paper or notepads during the consultation. Often, this information is not retained after the contact. In addition, SSCs have to interact with several dissimilar and often redundant systems and conflicting databases to issue a service order.

In this environment, system availability largely controls the length of an order transaction. SSCs input data, retrieve data, and interpret records from Billing Order and Support System (BOSS), Loop Maintenance Operation System (LMOS), Facilities Cable and Repair System (FACS), Premises Information System (PREMIS), Customer Appointment Management System (CAMS), Service Order Processing And Distribution (SOPAD), Service Order Negotiation And Retrieval System (SONAR), Service Control And Tracking System (SCATS), and other systems. They also have to consult with their customers concurrently—finding out their needs and recommending appropriate products to them. These activities and processes pose a heavy load on their cognitive resources and tend to interfere with the consultations with their customers. The problem is particularly pronounced for new SSCs.

The current service order entry systems do not effectively support SSCs or allow them to process most service orders while online with their customers. SSCs use these current systems to access current customer information. They use BOSS and Customer Account Retrieval System (CARS) to perform address validation and telephone number assignment.

SONAR was the first attempt to provide a system that allowed online negotiation. SONAR is the front end for various systems used in the negotiation of a service order. SSCs select the order type, and SONAR presents a screen flow based on the nature of the customer interaction—for example, a new connect or change order.

The SONAR user interface has not provided effective support for complex negotiation scenarios. The character-based screens, arranged in a predetermined sequence, do not support rapid changes in information content and flow during the course of service order negotiation. In addition, SSCs often find that an order will not flow through the system, requiring them to back out of SONAR and manually capture the order information.

As a result, SSCs resort to using native support systems during the negotiation process on a portion of customer requests, capturing pertinent customer and service information on note pads, and then preparing the service order after completing the call with the customer. The more lines being negotiated, or the more complex the service, the greater the likelihood that the SSCs prepare the orders manually. When preparing an order manually, SSCs do one of the following after completing the customer contact:

- enter a SONAR session and prepare the order
- enter the order information directly into one of the service order processing systems—for example, Customer Order Retrieval System (CORD), Service Order Logistic and Reference (SOLAR), or Service Order Processing and Distribution System (SOPAD)
- pass the service order form to an order entry clerk for input into one of the service order processing systems.

Installing the Intelligent Work Station (IWS)-based Computer-Aided Support System (COMPASS) began the process of making SSCs more productive. COMPASS allows concurrent access to multiple 3270 terminal emulations for host-based systems. This allows SSCs to multitask their access to various support systems. For example, SSCs can enter transactions into BOSS or SONAR and concurrently access PREMIS.

In their normal consultative sales tasks, the SSCs of the channels with a deployed sales helper will invoke CASH to profile their customers and recommend solutions to them. Recent usability studies of the Market Place Channel showed a combined CASH utilization rate of 56% (profile only, as well as profile and recommendation) in calls resulting in issuing of service orders. CASH utilization dropped substantially to 9.3% in calls that did not result in service orders.

One of the major bottlenecks identified is the lack of interface between legacy systems and CASH. In the Home Office Channel, all the service orders are generated through CASH on paper, then passed to order entry clerks for input into one of the service order processing systems. Market Place SSCs handle disconnects and simple change orders—for example, change of ring cycles—directly by using SONAR. Otherwise, the service orders are handled in the same way as in Home Office Channel.

In the usability studies of CASH currently deployed in the Home Office Channel (Phoenix) and the Market Place Channel (Omaha), the SSCs noted that the generation of paper service orders through CASH and the flow of such paper orders from the SSCs to the order writers are inefficient and error-prone.

3.2 Users

Targeted users are Mass Market SSCs. User interface design assumptions include that the age, gender, and disability distributions reflect the age, gender, and disability distributions of the adult population of the regions served by U S WEST. In addition, the SSCs are assumed to be fluent in English.

SSCs have a positive attitude toward customer contact and meeting customer needs, and a neutral attitude toward data entry. They are moderately motivated to perform data entry tasks.

SSCs have a good knowledge base. They know how to:

- sell U S WEST products and services
- interpret and produce service request
- respond to customer inquiries and complaints
- type using a standard QWERTY keyboard.

SSCs have some system experience. They currently use a workstation using IWS with 3270 (block mode) terminal emulation windows. They can input and retrieve data from Billing and Order Support System (BOSS), Customer Record Information Systems (CRIS), LMOS, Trunks Integrated Record Keeping System (TIRKS), and others. In addition, some SSCs have PC experience.

3.3 Usage Estimates

Most SSCs will use CCOE daily. Nonetheless, users may access parts of the system only occasionally, since some order activities and products are more commonly encountered than others. Therefore, the system must support users who do not have a great deal of experience and training in using it successfully.

A Small Business Group (SBG) study of ways to reduce distracting tasks for the SSC estimates system usage at 40 to 60 percent of an average 8-hour work day.

According to the informal usage estimates given by the subject matter experts (SMEs), there will be an average of 300 to 325 SBG SSCs (out of 600-650) and about 1600 H&PS SSCs (out of about 4000) accessing the Sales Helper at one time. In the future, there may be 9000 SSCs combined, and 40% of them will be accessing the Sales Helper at one time. In addition, BGS (Big Business and Government Service), Carrier, and other channels or market units may also be using CCOE in the future. The usage estimates will be revised as more information is available.

Formal usage estimates still need to be determined for Mass Market SSCs.

3.4 Usage Environment

The typical SSC work environment consists of cubicles arranged around a computer workstation. The phone is generally to the left of the workstation and most SSCs wear headsets to communicate with the customers. Many SSCs attach quick reference job aids to the walls of the cubicle.

Generally, the mouse pad and mouse sit to the right of the keyboard. However, some SSCs learn to work the mouse with their left hand so that they can write notes while moving the mouse.

The office is usually noisy. Noise sources include line printers, other SSCs talking on the phone, and SSCs talking to each other or to other U S WEST employees.

3.5 Hardware Environment

The user interface is designed for the standard Distributed Technology Platform (DTP) workstation. Workstation assumptions include support for the X Window System and OSF/Motif, and a 19-inch screen. The input devices are assumed to be a three-button mouse and a QWERTY keyboard with 24 programmable function keys as well as fixed function keys for cursor movement and editing functions such as home, end, page up, page down, insert, and delete.

3.6 Concurrent Applications

SSCs will use DTP workstations to access CCOE applications concurrently with other host sessions. They can access a combination of sessions concurrently with CCOE, limited only by the capabilities of the existing workstations and the current desktop—that is, IWS.

Users currently may establish as many as eight concurrent sessions, using any combination of the following systems. The more commonly used systems include BOSS/CARS (sometimes two windows), SOP (SOPAD,

SOLAR, CORD), PREMIS, Document Management Platform (DMP), and CAMS. The less commonly used systems include LMOS, FACS, and others specific to unique user requirements.

4. Design Conventions

This chapter discusses the CCOE design philosophy and presents the standard screen and window components and user inputs that support the philosophy.

4.1 Design Philosophy

Participatory and user-centered design have been used in the process of developing Sales Helper and related applications. Frequent usability evaluations have been performed with iterative design and prototyping of the user interface. Thus, a large degree of reusability and systematization of the design and contents of the system can be achieved. In the development of TIPS/CSPE, profile items from different but related projects such as TIPS/SSNQ (SBG), CASH (Home Office) and Market Place for Priority Channel have been reused, substantially reducing the time and effort of development. Joint development sessions were also held to foster exchanges of ideas and allow mutual validation of knowledge and design.

The most important design aspect of the CCOE system is its reliance on user-centered design principles. User-centered design requires, and is based on, user involvement during the entire development process. With this in mind, users are involved at every stage of design for the CCOE application. User participation is comprehensive and is not limited to the end evaluation, which is intended to uncover any residual user-interface design faults. An important aspect of user-centered design is iterative evaluation. Iterative evaluation involves developing prototypes of the system, which undergo testing, are changed, and evaluated again (one iteration). This process is continued until usability problems have been eliminated.

The TIPS/CSPE system is intended to support and facilitate consultative sales without constraining the SSCs to pre-determined sequences of interactions with the customers. The users are in control of the system, and the interactions are customer-driven—not system-driven—as no system can predict the course of conversation with the customers. In this way, the user can interact with the system driven by the conversation with the customer. It may be difficult to achieve total customer satisfaction without designing a consultative sales system driven by the consideration of customer needs.

The GUI design is based on the *OSF/Motif Style Guide Revision 1.2* and *U S WEST User Interface Overview for OSF/Motif*.

The user interface design philosophy:

- supports the tasks performed by the SSC
- reduces expenses by reducing system training time and reducing the need to use off-line reference materials
- improves customer service by reducing error tracking and corrections, improving order status accuracy, and reducing system interaction time
- increases revenue.

To achieve the CCOE user-support goals, the user interface incorporates the following characteristics. It:

- has a user-centered focus rather than a system-centered approach
- supports multitasking for the tasks that SSCs perform today
- employs an object-action metaphor; that is, users select graphic objects—for example, a customer—and specify an action to be carried out on the object—for example, display customer information
- uses windows and dialog boxes to display and solicit information
- supports interaction through both the keyboard and mouse
- requires a minimal number of actions to complete an order
- allows users to have direct access to as many parts of the consultative sales and order-entry process as possible, within the data base and hardware/software constraints
- allows users easily to navigate through windows and to make changes
- allows users to access detailed information in multiple ways
- provides progressive disclosure of status
- minimizes the need to type in information through use of menus, list boxes, autopopulation of fields, and techniques for copying information and specifying parameters in bulk
- detects errors
The application checks to make sure required data have been provided and to ensure validity of syntax and format of data, field content, and field cross-dependencies.
- provides understandable error messages in a timely fashion, allowing the user to find and correct errors easily
- allows the user to save an incomplete order and return to it later.

4.2 Screen Formats

The CCOE user interface employs the X Window System protocol, which conforms to the OSF/Motif window manager standards. The user interacts with the CCOE application using windows. A window is a rectangular area of screen that provides the user a functional means to communicate with an application and through which the application can communicate with the user. There are two types of windows in the CCOE application: main (primary or application) windows, and secondary windows or dialog boxes.

Applications appear in a main window, to which the user may apply most Motif window management functions. This application window spawns other windows for information display and data entry. The application windows cannot overlap the screen area that displays the IWS icons. All window layouts support a work flow that moves from the upper left of the window toward the lower right.

Panes are rectangular areas within a main window. They are visually demarcated, and they appear and disappear as a whole based on user commands—for example, by clicking on an icon or push button. The user cannot move a pane. Using panes ensures that the same information always appears in the same place on a primary window so the user can find it easily. Different panes may occupy the same area of a primary window at different times.

Secondary windows or dialog boxes are temporary windows that overlap, or partially overlap, a main window. The user may display more than one secondary window or dialog box at a time. The user can move secondary windows and non-modal dialog boxes to reveal displayed information beneath them. Modal dialog boxes, which cannot be moved, require a response to close the dialog box. Users must close a modal dialog box to continue working.

Minimizing or closing a primary window also minimizes or closes any associated open secondary windows or dialog boxes.

4.2.1 Main Window

Once opened, a main window remains open on the desktop until the user minimizes or closes the window. The CCOE application window is a main window. The application window controls other main windows, as well as secondary windows and dialog boxes. Closing the primary application window closes the CCOE application on the IWS desktop. A main window (Figure 1.) includes the essential components described after the figure.

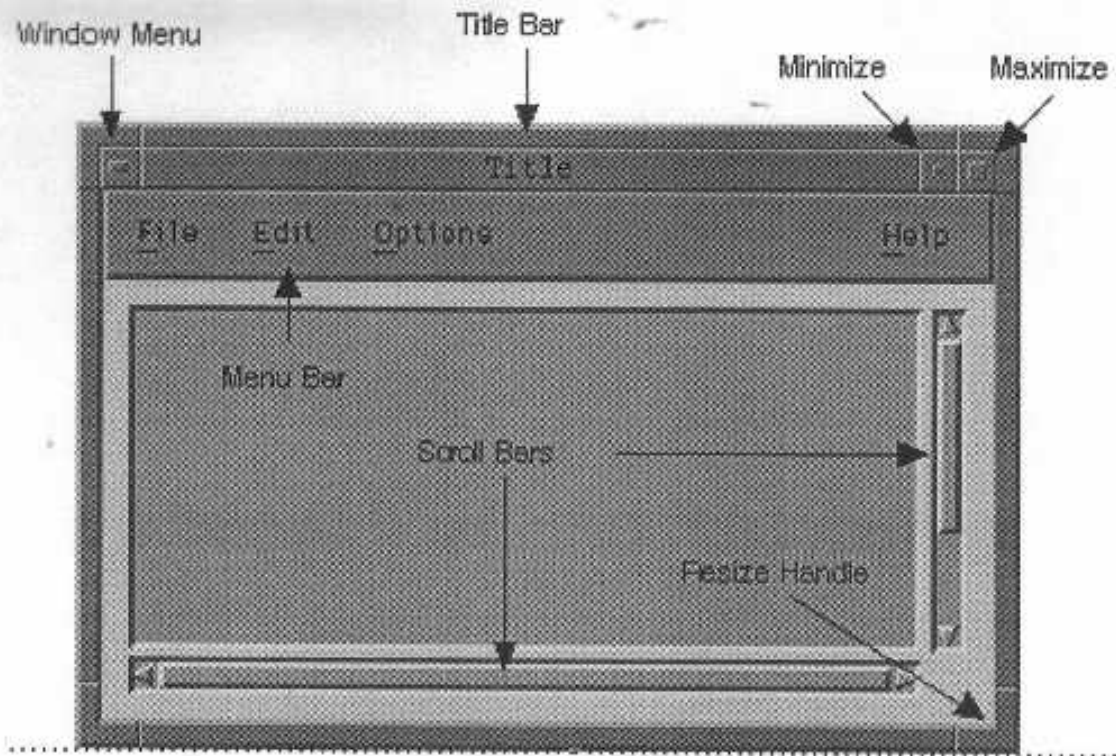


Figure 1. Main Window

Window menu: The window menu, a standard feature of Motif, provides window manipulation capabilities such as Move, Resize, and Close. The window menu allows users to manipulate windows with their keyboard rather than mouse. Double-clicking on the window menu button closes the window.

Title bar: The title bar contains the name of the window.

Minimize mutton: The minimize button allows mouse users to change the active window into an icon. Only a main window contains a minimize button.

Maximize mutton: The maximize button allows mouse users to expand the size of the main window to the maximum size permissible. Only a main window contains a maximize button.

Menu bar: The menu bar displays pull-down menu options that provide control and select functions such as File, Edit, and Help within the window. Only a main window contains a menu bar.

Scroll bars: Scroll bars appear in a window only when there is more information than can be displayed in the window frame. Scroll bars allow users to display hidden information. The CCOE main window contains horizontal and vertical scroll bars. The users can click on the directional arrows or drag the slider to scroll horizontally or vertically.

Resize handle: Resize handles are found in all four corners of a window. Mouse users can use the resize handles to resize a window quickly.

4.2.2 Secondary Windows or Dialog Boxes

The application uses secondary windows or dialog boxes to communicate with the user. Secondary windows and dialog boxes are information exchanges between the user and the application. They are used to perform specific functions that are too complex to be done with one of the standard controls or displays alone. The user must choose an option or press a button to send a response back to the application. Dialog boxes are removed when the user sends a response back to the application—for example, by pressing the OK button or Cancel button. Secondary windows and dialog boxes are always associated with a specific window and are displayed on top of that window. For CCOE, some dialog boxes must be acknowledged or responded to before any other actions can take place. For example, users must respond to critical system messages before other actions can be carried out.

Secondary windows and dialog boxes (Figure 2.) have a frame similar to that used by the main window and are usually made up of several of the standard interface components discussed in the next section.

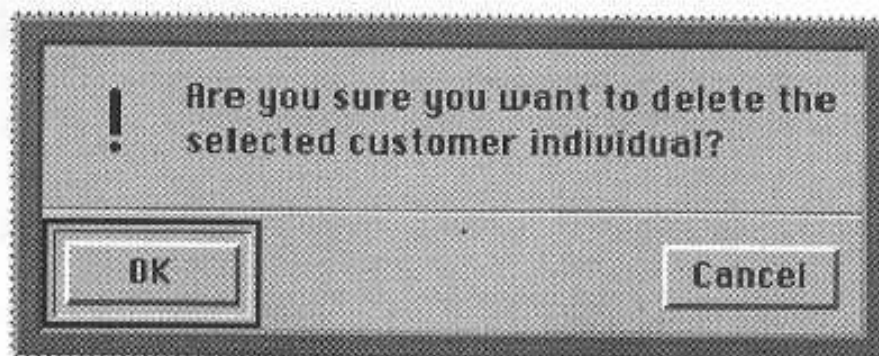


Figure 2. Dialog Box

4.2.3 Standard Interface Components

The CCOE application uses standard components throughout the user interface. These components comply with guidelines in the *OSF/Motif Style Guide Revision 1.2*, the *U S WEST User Interface Design for OSF/Motif*, and

conventions given in *Visual Design with OSF/Motif*. Standard components used in the CCOE application and their associated behaviors are described below.

4.2.3.1 Labels

Labels (Figure 3.) are used primarily for field or button names. When labels are used with a text entry field, the label is placed to the left of the entry field.

4.2.3.2 Text Entry Field

A text entry field (Figure 3.) allows data entry and displays the current contents of the data base field. Users can type new text, or insert, delete, or change the text in a text entry field. In other words, a text entry field supports basic editing functions. The text cursor must be positioned in the entry field before the user can start typing. When the cursor is in the text entry field, the borders of the field turn black.

CCOE validates whether users have entered permissible data in text entry fields when possible. For example, if users enter alpha characters in a numeric text entry box, the system informs the user that the entry is not valid.



Figure 3. Text Entry Field with Label

4.2.3.3 Scroll Fields

A scroll field (Figure 4.) displays lists or strings of continuous text. Users can scroll these fields to scan through the contents. If the list or text extends beyond the vertical or horizontal boundaries of the field, the scroll bar may be used to traverse the information.

If the scroll field is used for displaying a list, each line represents a list item and may be selected. If scroll field list items are numbered, options range from 1-n. They do not begin with zero. Parts of a line in a list may not be edited or selected directly.

Unless otherwise stated, selection of list items is handled as follows:

- Clicking on a line highlights and selects it.
- Clicking on a line and dragging the cursor across adjacent lines with the mouse button depressed highlights and selects those lines when the mouse button is released.
- Holding down a **<Control>** key while making the selections allows discontinuous selection of items or sets of items in a list. All items highlighted are selected, even when they are not contiguous.

If the scroll field is used for text, the standard editing functions may be applied.

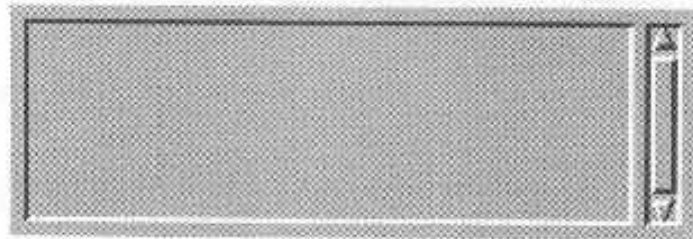


Figure 4. Scroll Field

4.2.3.4 Pull-down Menus

A pull-down menu (Figure 5.) lists the options available from cascade buttons on the menu bar in main windows. When the user selects a cascade button from the menu bar, a pull-down menu of options appears. The user can select an option from the menu by dragging the mouse and releasing the mouse button on a particular element. The user can also select options from the keyboard. A mnemonic—for example **<Alt>+ <F>** for the File menu—provides a quick way to access a menu option from the keyboard.

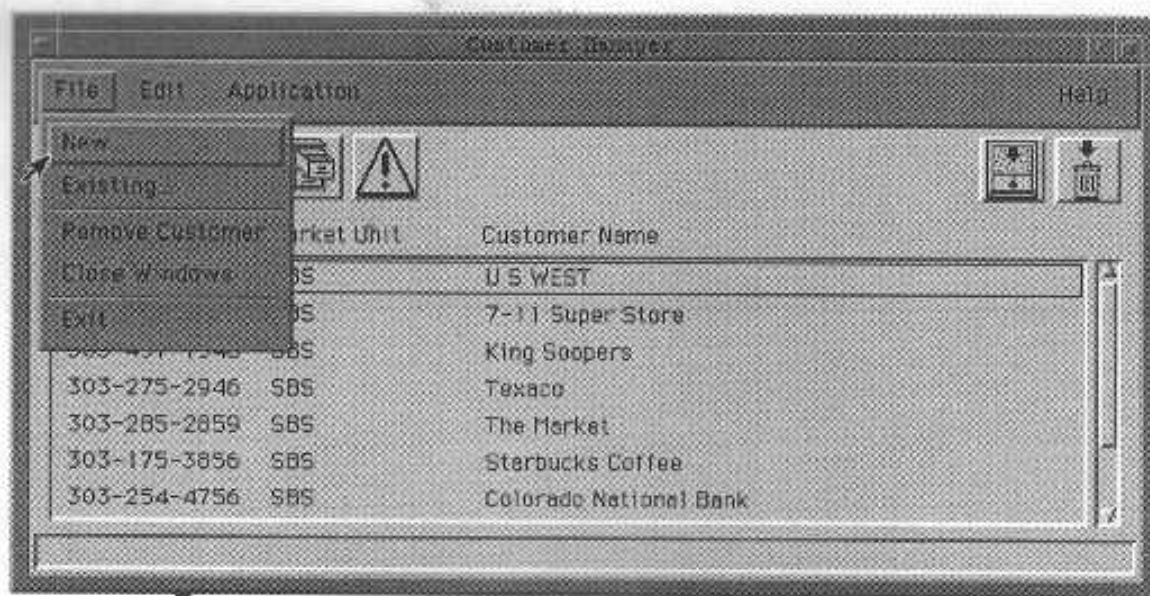


Figure 5. Pull-Down Menu

4.2.3.5 Option Menu Buttons

An option menu button (Figure 6.) displays a single element (option) from a menu. The small rectangle on the right side of the button label indicates that it is an option menu button. Selecting the option menu button displays the entire option menu. The user may select one element from the menu, which then closes. The newly selected element replaces the previously displayed element on the option menu button.

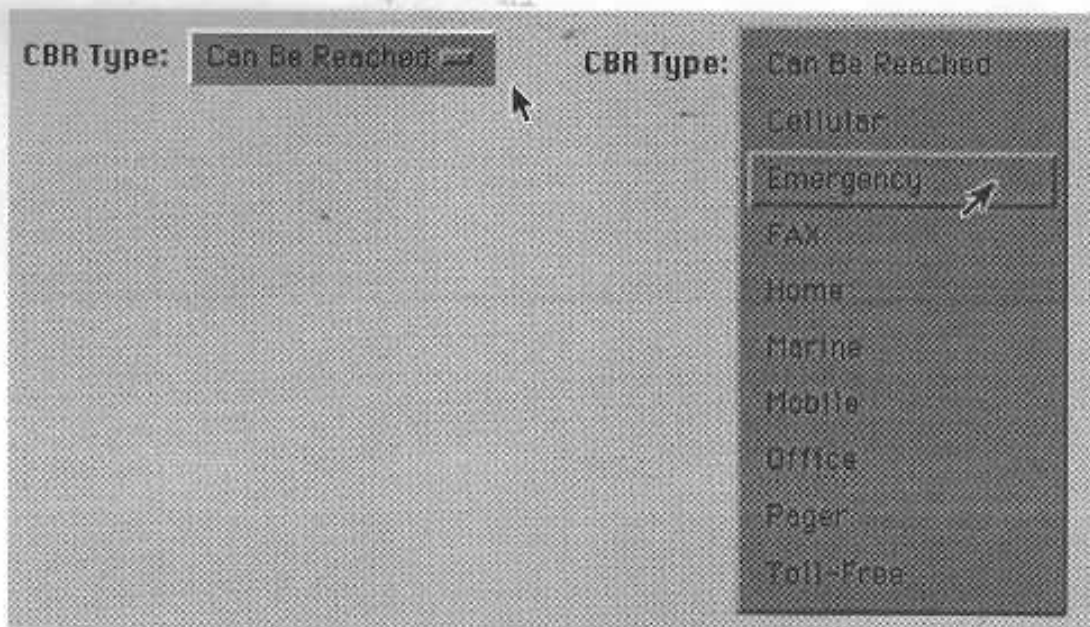


Figure 6. Option Menu Button and Option Menu

4.2.3.6 Drop-down Combination Box

A drop-down combination box (Figure 7.) combines a text entry field with a selectable scroll field. A user either types information into the text entry field or populates the entry field by selecting an item from the scroll field. Only the text entry portion of a drop-down combination box is visible until the user selects the arrow push-button to display the scroll field portion. Drop-down combination boxes are not supported on all platforms.

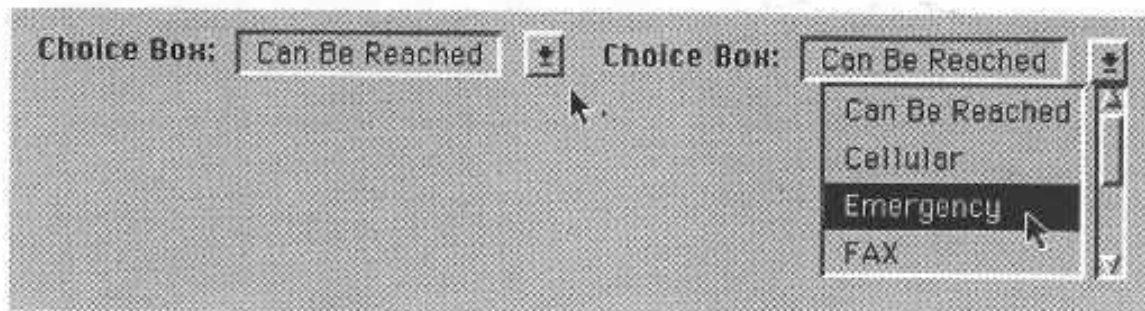


Figure 7. Drop-Down Combination Box and Scroll Field

4.2.3.7 Command Buttons

A command button (Figure 8.) initiates an action. Actions may include displaying a dialog box, confirming a choice, or starting a desired process. Command buttons are rectangular boxes which typically appear at the bottom of dialog boxes. Command buttons may also appear in vertical columns located on the right side of the display area. Each command button contains a label with the command name. Users must select the command button to invoke the command.

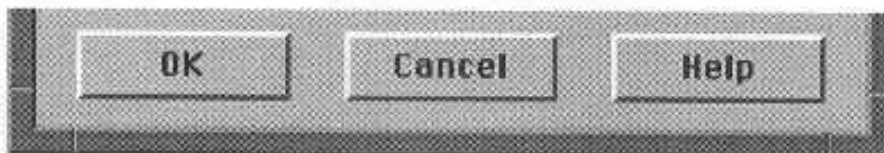


Figure 8. Command Buttons

These buttons are common across applications. Activating them triggers the following actions:

- OK causes the application to apply any changes and perform related actions specified by components in the dialog box, then dismisses the dialog box.
- Cancel closes the dialog box without performing any dialog box actions not yet applied to the application.
- Help accesses online help information for the dialog box.

4.2.3.8 Check Boxes

Check boxes (Figure 9.) are square buttons with labels. Check boxes allow the user to select one or more items in a group of options. A selected button appears filled. Reselecting on a selected button undoes the selection and returns it to its empty state.

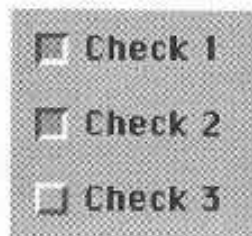


Figure 9. Check Boxes

4.2.3.9 Radio Buttons

Radio buttons (Figure 10.) are diamond-shaped buttons with labels. Radio buttons display a group of options that are mutually exclusive. A selected button appears filled. Reselecting the selected button undoes the selection. Selecting another button in the group selects that button and undoes the previous selection.

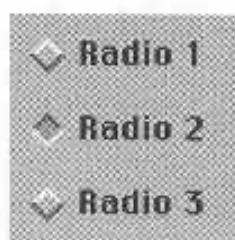


Figure 10. Radio Buttons

4.2.4 Tri-State Toggle Buttons

Tri-state toggle buttons provide a custom interface component. In the Sales Helper module of CCOE, it is necessary to represent the three states [yes | no | unknown] of the tri-state profile items. A group of two toggle buttons is used to represent each item, with the yes button on the left and no button on the right. The state transition diagram in Table 1. details the behavior and action for the state changes. It is followed by a representation of the buttons in Figure 11.

Original State to Final State	Action	Behavior
unknown to yes	click on yes button	yes button toggles on
unknown to no	click on no button	no button toggles on
yes to no	click on no button	yes button toggles off; no button toggles on
yes to unknown	click on yes button	yes button toggles off
no to yes	click on yes button	no button toggles off; yes button toggles on
no to unknown	click on no button	no button toggles off

Table 1. State Transition Table of Tri-State Toggle Buttons

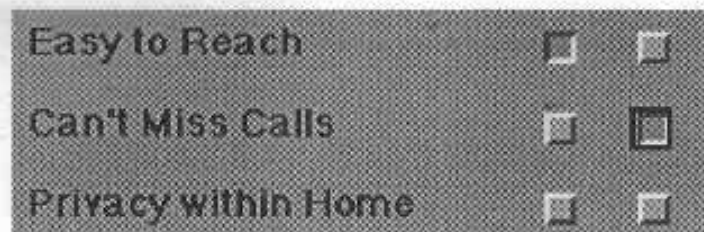


Figure 11. Tri-state Toggle Buttons

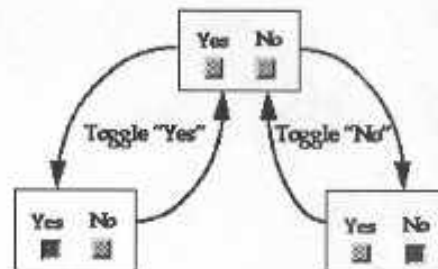
Figure 12. illustrates the three permissible states, the forbidden state, and the state transitions for tri-state toggle buttons.

State	Representation of Interface Objects	
Unknown	Yes	No
Yes	Yes	No
No	Yes	No

Allowable States

Yes No

 Forbidden State



State Transitions between "Unknown", "Yes" and "No"

Figure 12. Tri-state Toggle Button States

4.2.5 Icons

An icon is a picture or a symbol representing an object, task, message, or concept. An icon represents a familiar metaphor or an underlying concept. Icons can display a great deal of information in a limited space. A set of graphic symbols can be made visually more distinctive from one another than a set of words. Consequently, it is easier to locate a graphic symbol among a group of symbols than to pick out a word among a group of words.

CCOE icons (Figure 13.) represent commonly accessed functions or information that SSCs want to access quickly. The icons used in CCOE Sales Helper are placeholders. Further design and art work needs to be done on those icons.

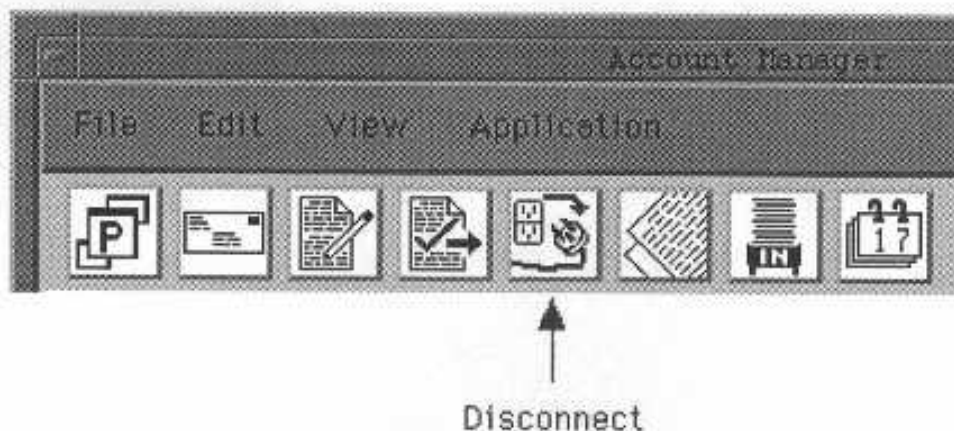


Figure 13. Icons

4.2.6 Use of Color

Color serves as a powerful visible cue that can draw a user's attention, set off an area on the user interface, or convey information to a user. Used properly, color can enhance the effectiveness and appearance of a user interface. Used improperly, it can be distracting and visually fatiguing, significantly reducing the usability of the interface. Research on the use of color in user interface design suggests that no more than seven discriminable colors be used in a display, including black and white. The benefits of color as an attention getter, information grouper, and value assigner are lost when too many colors are used.

Color can serve to:

- draw attention to important information
- provide status information
- identify different screen components
- increase screen appeal
- portray logical groupings of ideas.

On IWS terminals, users can customize the colors for different applications. However, color selections are highly subjective, and some customized combinations do not support good performance. Given this, the CCOE

application offers a choice of several color combinations for users. These color combinations are designed to maximize readability and minimize eye fatigue.

4.2.6.1 Color Coding

Color perception may vary substantially between individuals. For example, older people are less sensitive to color and require higher brightness levels to discriminate among colors. Roughly 9% of males and 5% of females have color-deficient vision and cannot utilize color codes in their range of deficiencies. Finally, cultural differences in color use and significance influence both the number of colors an individual recognizes and the meaning those colors. In most Western societies the color red indicates "warning," the color yellow indicates "caution," and the color green indicates "go." The CCOE application maintains consistency with the color associations identified for Western societies and summarized in Table 2.

Color	Symbolize
red	danger, failure, fire, hot, stop, unsatisfactory condition, warning, malfunction
yellow	caution, delay, slow, attention
green	acceptable, clear, go, ok, ready, safety, satisfactory conditions, vegetation
gray	neutrality
white	neutrality

Table 2. Conventional Meaning for the Symbolic Use of Color

In the Sales Helper module, a color coding scheme provides visual cues for follow-on questions; for next or previous page indicators for different page follow-on questions; and for common solutions having different capabilities—that is, the same solution can satisfy different needs of the customers. The color coding scheme is summarized in Table 2.

Color	Symbolize
green	follow-on questions, next-page indicator for follow-on questions
TBD (salmon)	a set of common solutions fulfilling multiple needs
TBD (sky blue)	a set of common solutions fulfilling multiple needs

Table 3. Color Coding Scheme for Sales Helper

Color	Symbolize
TBD [turquoise]	a set of common solutions fulfilling multiple needs
TBD	additional sets as required

Table 3. Color Coding Scheme for Sales Helper

4.2.7 Fonts and Point Size

CCOE uses standard fonts recommended in the *US WEST Desktop and Window Management Environment for OSF/Motif*. Guidelines recommend using a proportional sans-serif font for on-line computer interfaces, except in word-processing applications where large blocks of text will be entered. **Helvetica** is an example of a proportional and sans-serif typeface.

Font and point size selection are based upon a standard viewing distance of 500 millimeters (20 inches). The default character size and font for CCOE is defined as **Helvetica Bold 14** for window titles and field labels, and **Helvetica Regular 14** for user input.

4.2.8 Alignment of Graphic Components

Left-align all text entry fields. To allow easy comparison of numeric values, right-align or decimal-align numeric text entry fields. Left-align all labels.

Align the choice label and selection indicator horizontally for check boxes and radio buttons. Left-align all labels. Position horizontally aligned buttons/labels enough apart that users do not accidentally select adjacent buttons/boxes. Keep buttons/boxes and labels consistent in appearance.

Align all entry fields and selection fields—for example, command buttons, check boxes, and radio buttons. Align the leftmost text entry field under the leftmost check box or radio button. Left-align all labels.

4.2.9 Headings for Graphic Components

Use window titles of no more than three words. Display the title in mixed case. Follow standard rules of capitalization for all labels.

4.2.10 Core Commands

The core commands conform to the *OSF/Motif Style Guide Release 1.2* and the *U S WEST Application User Interface Design for OSF/Motif*.

4.3 User Inputs

SSCs use a three-button mouse and a keyboard to communicate with the CCOE application. Input methods, accelerators, and tabbing sequences comply with the *OSF/Motif Style Guide Revision 1.2* and the *U S WEST Application User Interface Design for OSF/Motif*.

Whenever a window or panel cannot respond to user input, the pointer appearance in that window or panel changes appearance as specified in the *OSF/Motif Style Guide Revision 1.2* and the *U S WEST Application User Interface Design for OSF/Motif*. If system response is delayed by more than ten seconds and users cannot continue to work during that delay, a message dialog box displays. The message dialog box contains an indicator of progress and allows the user to cancel the operation.

Use focus—the location of the active entry cursor—to show the user the active field.

4.3.1 Mouse Conventions

The primary means for user input are a three-button mouse and the keyboard. CCOE requires the use of only the first, or left, mouse button (right handed mouse). Users can use the mouse to perform all functions and operations except typing textual data.

4.3.2 Keyboard Mapping

Provide keyboard equivalents for all mouse actions. Unless otherwise indicated, user inputs used in CCOE conform to the conventions established in the *OSF/Motif Style Guide Revision 1.2* and the *U S WEST Application User Interface Design for OSF/Motif*.

Define the keyboard equivalents as mnemonics—**<Alt>** + the key indicated— or as accelerators—**<Control>** + the key indicated.

4.4 Wait State

The core commands conform to the *OSF/Motif Style Guide Revision 1.2* and the *U S WEST Application User Interface Design for OSF/Motif*.

5. Business Functions and Processes

Although there are various identified stages of the consultative sales contact—rapport-building, interviewing, presenting, and closing—neither its sequence nor the profile items that will be visited can be predicted. It depends on the type of contact, the personalities of the SSCs and the customers involved, and the unique situation of that particular contact. In the design process, we have attempted to group related profile items to natural categories and to put them in a typical sequence of contact.

Basically, the profile collection process roughly corresponds to the rapport-building and interview stages of the contact. The summary and recommendation process corresponds to the presentation of solution and closing stages by providing pertinent information that relates the needs of the customers to the recommended products and the product benefits to the customers.

There is no known task flow for consultative sales except for a few broad stages like establishing rapport, interviewing, presenting, and closing. Naturalistic observation, interviews, and focus groups all confirm that there are no well-defined methods or procedures to conduct consultative sales. The purpose of the Sales Helper system is to provide the right information to the user at the right time with user-centered control. The work flow diagrams (Figure 14. and Figure 15. in Section 5.3) indicate that the overall flow is driven by the customer—that is, the transition to recommendation and whether more interview is needed—rather than by the system. The design of the flow of the Sales Helper is compatible with the flow of contact.

Although there is no pre-determined flow of conversation in many customer contact situations, the Sales Helper environment is modelled after the natural order of the profile items and the maximal probability of having the follow-on on the same page of profile items.

5.1 PMO Task Analysis

A task analysis describes, from a user perspective, the tasks that the user performs to accomplish major business functions. It breaks down each major business function into its component tasks. These tasks can be further broken down into the inputs and outputs the users require to operate the system efficiently. Reengineering a job and its accompanying processes results in new processes. However, the PMO is the only task analysis that can be verified. PMO task analysis provides a starting point from which to incorporate new business functions, data, and processes into the system interface that must reflect the PMO.

Extensive task analyses have been completed for the Mass Markets. These task analyses represent the PMO for both HPS and SBG. They provide important information concerning the data and processes that constitute the core of the FMO or, at the very least, the transition from PMO to FMO. In the absence of documented FMO processes, they represent one of the richest sources of information for the UI designer.

5.2 New Business Functions

The CCOE Sales Helper represents a new business function, one not available in PMO. CCOE enables the SSC to gather pertinent profile data about the customer site and to store that data in the CSDBs. This allows the SSC to retrieve information about existing customers. The CCOE Sales Helper recommends solutions based on customer needs and profile information. The reasons why the products are recommended by the system and the benefits of the products are also made available.

In the legacy systems negotiation and ordering environment, this information is generally not gathered or available, even though the SSCs have to make use of this information all the time to make recommendations to the customers.

With CCOE Sales Helper, the SSC can create, view, update, and delete customer site profile information for new or existing customers for Mass Market. Additionally, the SSC can make use of the recommendation and the information provided as a sales aid in their consultative sessions with the customers.

The CCOE Sales Helper supports the following major business functions:

- creating, displaying, and updating customer site profile information
Table 4. summarizes the SBG and H&PS information categories.
- displaying the date of last contact
- displaying the current known values of the customer site profile
- providing indicators for follow-on questions (profile items) relevant to the course of the contact
The knowledge-based system and associated user interface will tag the questions according to the state or value of the current profile item selected.
- providing a help definition and possible consultative questions for each profile category
- providing a help definition for each profile item

- generating customized recommendations based on the customer site profile
The recommendation is organized in terms of possible solutions under the capabilities of the solutions to satisfy the customer site profile.
- providing need traces that relate the capabilities of each group of solutions to the customer site profile items
- providing solution benefits for the recommendation offered
- providing access to a product ordering module through a proposed product list window and/or direct link.

The Sales Helper Profile windows support these functions. The *Customer Contact Operating Environment Sales Helper Functional Specification* will further describe these functions and their relationship to the corporate logical data model. The functional specification will also provide a table to map entities and attributes to interface windows and fields.

SBG	H&PS
business information	service status
motivation of the call	home layout
layout of the business	motivation of the call
problems and concerns experienced	telecommunication equipment
business needs for easy telecommunication	usage information
business needs to identify calls	phone users information
business needs to control calls	life style
the answering method used	concerns
needs of any mobile workers	customer needs
line usage information	
usage patterns	
telecommunication equipment	
image the business wants to project	
visibility the business will like to have	
type of advertising	
type of business clientele	

Table 4. Customer Site Profile Information Categories

5.3 New Tasks and Workflow Processes

The user interface presents a new process and a series of tasks that accomplish these new business functions. These tasks manage new customer data in a manner that reflects the customer entities, their attributes, and the relationship between entities as specified by the CSDBs. Because the tasks (Figure 14. and Figure 15.) are designed to be tightly coupled with the interface windows and the data each presents, the most faithful representation of tasks in a reengineered system is a work flow process that includes references both to task windows and to the tasks they accomplish.

For a detailed interface specification of each tasks referenced in the figures, see section 6. Sales Helper Interface Specification. The subsections discussing each window include Control/Action/Response (CAR) diagrams. These diagrams provide information for each control on each window, describing the appropriate user action and the resulting system response to that user action.

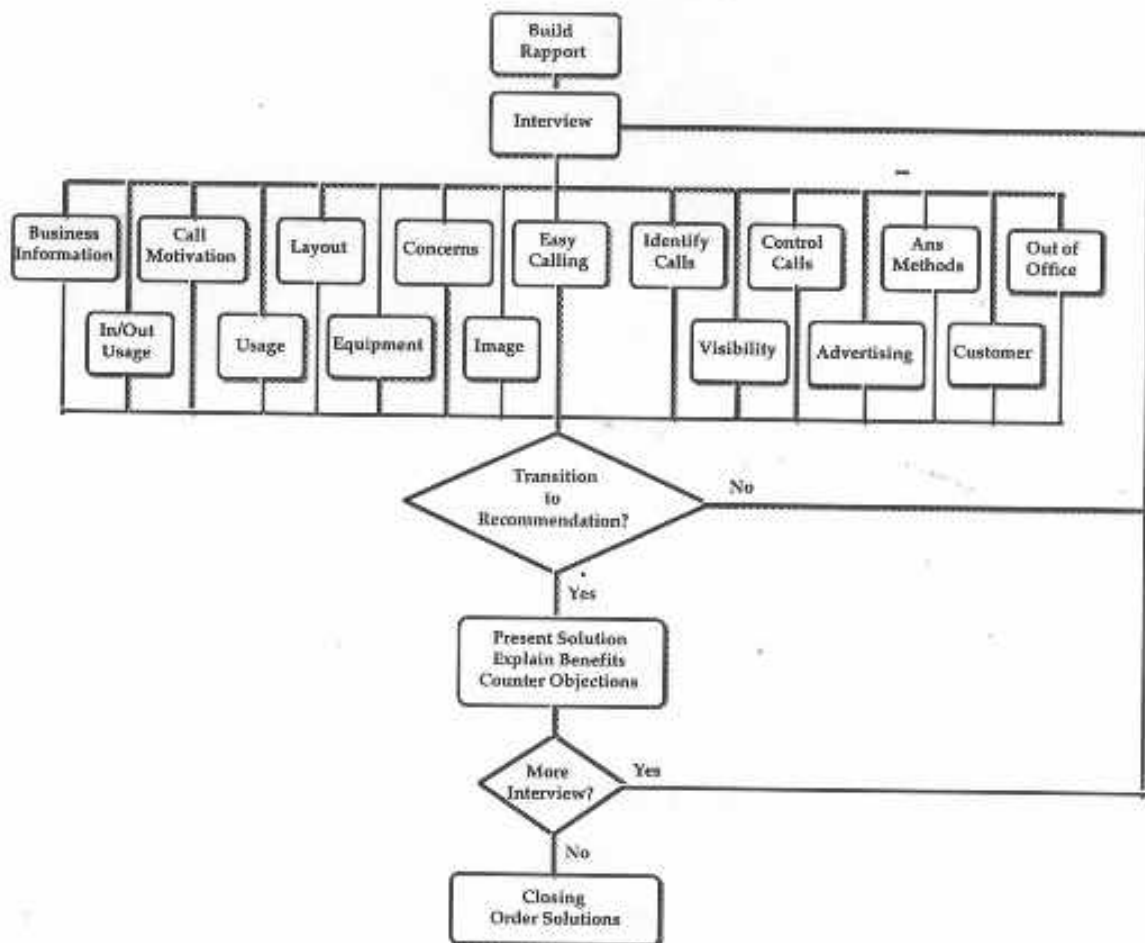


Figure 14. Workflow Process of consultative sales (SBG)

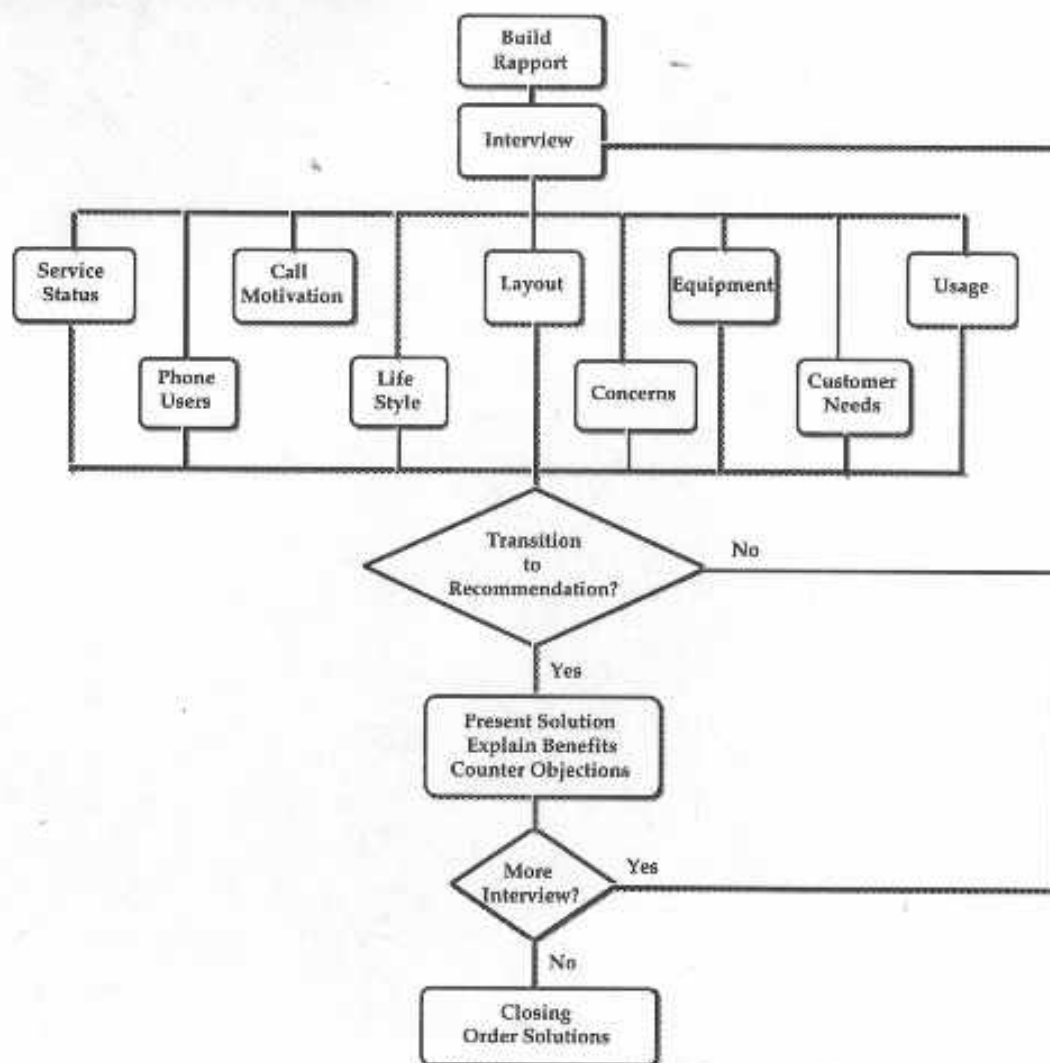


Figure 15. Workflow Process of consultative sales (H&PS)

5.4 Functional Organization of the Interface

The CCOE Sales Helper (Figure 16.) controls all the functions defined by the consultative sales process at the customer site level. The Sales Helper main window and its associated windows retrieve customer site profile information records, display customer site profile information, and allow SSCs to create and update the pertinent information. The Recommendation Window displays possible recommended solutions organized in capabilities. The product manager can also be accessed through the recommended solutions to simplify the negotiation and ordering process.

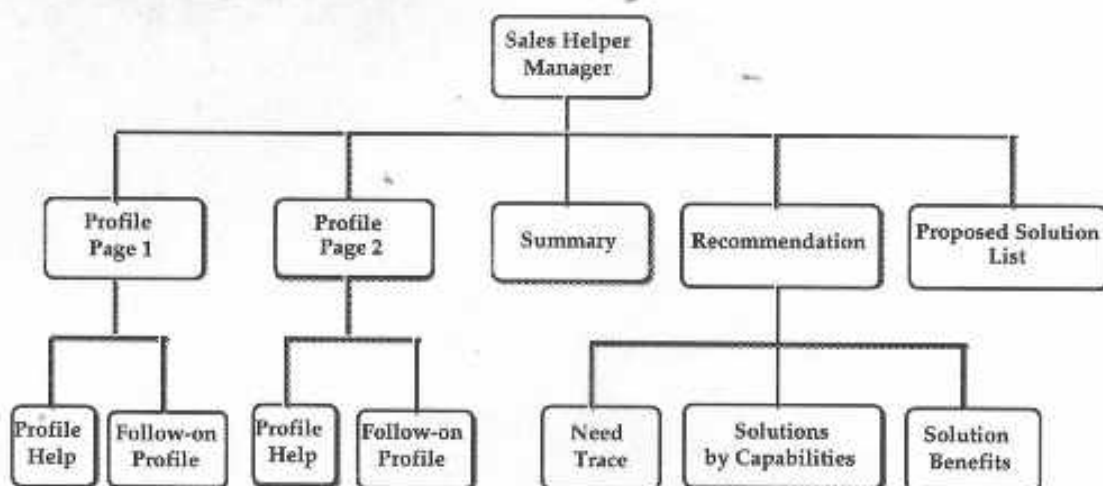


Figure 16. Functional Organization of User Interface of Sales Helper

6. Sales Helper Interface Specification

This chapter contains:

- a brief overview of Sales Helper
- Sales Helper business functions in section 6.1
- Sales Helper general description in section 6.2
- Sales Helper window specification in section 6.3
- Profile pane description in section 6.4
- Profile business function in section 6.5
- SBG Profile panes description and specification in section 6.6
- HPS Profile panes description and specification in section 6.7
- Profile Summary pane description and specification in section 6.8
- Recommendation window specification in section 6.9 .

Before invoking the Sales Helper, the SSC must have determined whether the customer is new or existing and must have identified the market unit of the customer site. The CCOE Customer Manager standardizes the name of the customer and validates it against the CSDB.

The SSC will open the Sales Helper from the Customer Manager or the customer product viewer. (Access from other modules—for example, repair—will be available in future releases.) The Sales Helper then makes available the profile item panes that correspond to the customer status and market unit. If the customer is new, the Sales Helper will display the first profile pane. If the contact involves an existing customer with an existing site, the Sales Helper will display the profile summary information associated with the site to facilitate building rapport with the customer.

It is not necessary for the SSC to specify a new customer site when initially collecting profile information. However, site profile information must be associated with a valid customer site for the Sales Helper to provide resolution recommendation or to save profile items to the CSDB.

For the initial phase of Sales Helper, customer site information will not be available, and the recommendation will be based the state served by U S WEST in which the service is provided. The mechanism for identifying the state is to be determined.

6.1 Sales Helper Business Function

The Sales Helper supports the following major business functions:

- entering new customer site profile information
- updating existing customer site profile information
- displaying a summary of the customer site profile information
- displaying customized recommendation of solutions meeting customer needs at the chosen site
- providing need/solution trace and product benefits
- providing definitions of profile items and categories
- providing examples of pertinent consultative questions.

See the *Customer Contact Operating Environment Functional Specifications* for a full description of the specification of these business functions.

6.2 Sales Helper General Description

The Sales Helper window (Figure 17.) opens when the application is launched from the Customer Manager, Customer Product Viewer, or other CCOE screens that will be developed. The menu bar on the Sales Helper window provides both pull-down menus and icons. The SSC selects from these menus and icons to accomplish various business functions. The Sales Helper main window makes several panes available, based on the context of the customer contact. The menu bar and icons correspond to the pane being displayed. Sales Helper panes include separate Profile Summary, Profile 1, and Profile 2 panes for each market unit; and a Recommendation pane that displays possible sets of solutions recommended by a knowledge-based system.

After entering the customer site market unit, the SSC then identifies whether the customer is new or existing. At that point, the SSC can choose either to enter new customer information or to use Sales Helper, whichever is compatible with the flow of conversation.

For a new customer, the SSC invokes Sales Helper with the New Customer icon or with the New Customer button on the Customer Manager menu bar File pull-down menu. Sales Helper opens with the Profile 1 pane for the market unit.

For an existing customer, the SSC searches for the customer and can invoke the Customer Product Viewer (or other functions such as updating credit information) in Customer Manager. Once a site is selected in the Customer Product Viewer, the SSC can invoke Sales Helper for that site at any time, and from any screen where Sales Helper is available. If the contact involves an existing customer with one or more known values for profile items

associated with the selected site, Sales Helper opens with the Profile Summary for the site. Otherwise, Sales Helper opens with the Profile 1 pane for the market unit. Figure 17. displays the HPS Profile 1 pane.

The screenshot shows the 'Sales Helper Window (H&PS)' with the 'Profile 1' pane. The window has a menu bar (File, Edit, View, Application, Help) and a toolbar. The main area is titled 'Customer' and contains several sections:

- Service Status:**
 - New Service: ☒ Yes, ☐ No
 - Moving: ☒ Yes, ☐ No
 - Total Deactivate: Unknown
- Layout:**
 - Home Type: One Level, Multi Level, Apartment, Trailer, Other, Unknown
 - Detached Lock: Unknown
- Call Motivated by:**
 - Media: Unknown
 - Promotion: Unknown
- Equipment:**
 - Phone Sets: Unknown
 - Ans Mch / Sec: ☐ Yes, ☐ No
 - Pager: ☐ Yes, ☐ No
 - Cellular / Mobile: ☐ Yes, ☐ No
 - Modem / Fax: ☐ Yes, ☐ No
 - In Local: Unknown
 - Out Local: Unknown
 - Toll In LATA: Unknown

At the bottom, there is a status bar that reads 'Structure of home including levels, living space, and work areas' and 'Last Update 5-25-1995'.

Figure 17. Sales Helper Window (H&PS)

The SSC typically begins the consultative session by establishing rapport with the customer and capturing customer site-specific information as well as customer needs. The SSC clicks on the buttons or chooses from the option menus of the profile items provided on the profile panes. For follow-on questions (profile items) pertinent to specific profile items, the background of the follow-on profile items line will turn green if the state of the item is unknown. Follow-on questions that fall on different pages will be indicated by turning the page button (next or previous) green. All the follow-on questions are optional, and the SSC can use them depending on the circumstances of the conversation. The flow of the customer contact guides the flow of information through the profile windows. By choosing the Summary Icon or the Summary button from the menu bar's View pull-down menu, the SSC displays the Sales Helper Profile Summary, which contains a summary of all the profile items with known states.

The SSC can choose the Recommendation icon or the Recommendation button from the menu bar's Application pull-down menu to invoke the Sales Helper Recommendation window any time during or after the interviewing process. The recommendation reflects the current choices of the profile items just before the recommendation button is pressed.

The Recommendation window consists of sets of recommended solutions generated from the knowledge-based engine and knowledge base of Sales Helper. The recommended solutions are organized in groups of capabilities that can satisfy customer needs or that match customer profile information. The SSC can view the need/profile item trace (linking capabilities to specific profile items) or solution benefits by moving the mouse over the corresponding item.

When the mapping from solution to product is one-to-one, the SSC can select a solution and put the corresponding product into a Proposed Products list window by single clicking on a solution. In the case of a solution having different flavors or choices, a dialog box offering the specific products will appear, prompting the SSC to select the products the customer may want to order. The order of presentation of the different flavors will be based on the SMEs' estimates of the frequency with which they are typically ordered.

The Proposed Products list window will be used to navigate around the product manager. Access alternatives include clicking on a product in the proposed product list and double-clicking on an item in the Recommendation window. The access method is to be determined.

Icons in the Sales Helper window allow users to go directly to other Sales Helper panes, the product manager, or Help without pulling down a menu from the menu bar. The icons are (left to right): Next (Previous) Page, Summary Page, Recommendation, Product Manager, Turn Help Off/On, and Help.

The Sales Helper window icons or menu commands accomplish the following tasks:

- Next/Previous Page displays the other of the two profile panes in the Sales Helper window.
- Summary displays the summary pane in the Sales Helper window.
- Recommendation displays the solution recommended by the underlying Knowledge-Based System in the Sales Helper window.
- Product Manager displays the Product Manager of CCOE. The SSCs will choose whatever products they need to visit by using the icons on the Product Manager.

6.3 Sales Helper Window Specification

The Sales Helper window specification consists of Table 5., which describes window behavior, a summary of any window edits, Table 6., which describes the window fields, and Table 7., which describes the control/action/response sequences for the window fields.

6.3.1 Sales Helper Window General Information

Item #	Description	Behavior
1	General window behavior	Sales Helper window opens when the application is launched from Customer Manager or Customer Product Viewer menu. If the SSC identifies the customer site of an existing customer before the application is launched, a summary of the existing profile information appears in the window. Otherwise, a new profile entry pane appears. The following panes can be accessed from this window: Profile 1, Profile 2, and Summary.
2	Window behavior: relationship to other windows	A separate recommendation window can be invoked by the recommendation icon (or recommendation under Application Menu). A proposed product list window is also available whenever the user selects a solution by clicking on it.
3	Parent geometry and placement	This window opens superimposed upon the Customer Product Viewer window, slightly below and to the right of its title bar.
4	Dialog and error message geometry and placement	Information and Error Message dialog boxes appear upon the window to which the message refers. They are centered horizontally on the window from which they were spawned, one-fifth of the window distance down from the top of the window.
5	Window behavior: color usage	Not specified at this time.
6	Window performance requirements	Opens within a second of selecting from the menu bar or icon.
7	Error messages	Error messages are displayed as a result of choosing the following icon or command: •Clear All to Unknown: Do you want to clear all the profile items of this customer site to Unknown? •Exit: Do you want to exit this application?
8	Reminders and disclosures	N/A
9	Help	Icon, profile category and profile item help messages will appear in the message area when the cursor is positioned on the corresponding widget. A complete list of the messages is included in the appendix. The definition and the description of the profile category and item will be provided by the knowledge base.

Table 5. Sales Helper Window General Information

6.3.2 Sales Helper Window Edits

No window edits have been identified at this time.

6.3.3 Sales Helper Window Field Descriptions

#	Field Name	Type	Field Length/Format	Field Description
1	File	menu bar cascade button	N/A	contains these options: •New Site •Existing Site •Save as Site •Exit
2	Edit	menu bar cascade button	N/A	contains these options: •Clear all to Unknown •Restore to Previous
3	View	menu bar cascade button	N/A	contains these options: •Profile Page 1 (or 2) •Summary
4	Application	menu bar cascade button	N/A	contains these options: •Recommendations •Product Manager
5	Help	menu bar cascade button	N/A	contains these options: •TBD
6	Next (Previous) Page icon	iconic push button	N/A	Pushing this button calls the Profile 2 (Profile 1) window.
7	Summary icon	iconic push button	N/A	Pushing this button calls the Summary window and dismisses the profile windows.
8	Recommendation icon	iconic push button	N/A	Pushing this button calls the Recommendation window showing the solution recommended for the customer site.
9	Product Manager icon	iconic push button	N/A	Pushing this button calls the Product Manager window.
10	Help On/Off icon	iconic push button	N/A	Pushing this button turns the multi-line help box in the window message area off and on. Default is on.
11	Help icon	iconic push button	N/A	Pushing this button calls the Help Manager.

Table 6. Customer Manager Window Fields

#	Field Name	Type	Field Length/ Format	Field Description
12	Customer Name	push button	N/A	TBD
13	Profile Page 1 pane	scrollable form	N/A	contains the profile categories panes inside which profile items are placed; market unit-specific. Only one of the panes (13,14,15) can be displayed at the same time. Details in CAR diagram and Appendix A.
14	Profile Page 2 pane	scrollable form	N/A	contains the profile categories panes inside which profile items are placed; market unit-specific. Only one of the panes (13,14,15) can be displayed at the same time. Details in CAR diagram and Appendix A.
15	Summary pane	scrollable form	N/A	constructed dynamically and displays the current states of known profile items. Only one of the panes (13,14,15) can be displayed at the same time.
16	Window message area	multi-line help area	N/A	displays text associated with individual category or profile item. The text includes item or category definition and typical consultative questions. It is activated by the position of the mouse pointer.

Table 6. Customer Manager Window Fields

6.3.4 Sales Helper Window CAR Diagrams

This section captures interface behavior as specified for each control on the Sales Helper window. The following table specifies each control on the Sales Helper window, the allowable user actions, and the corresponding system response associated with each of these controls.

For user behavior, the following notation is used:

click = user clicks on item (button) using mouse button 1.

<Tab> = user tabs to item (button) and then <Return>.

mnemonic (key) = user activates mnemonic by <Alt> and the key indicated in parentheses.

accelerator (key) = user activates accelerator by <Control> and the key indicated in parentheses.

Control Area	User behavior	System response
Control Area I: WINDOW MENU BAR		
File cascade button	click, <Tab>, mnemonic (F) or accelerator (F)	spawns a pull-down menu displayed below File cascade button
File>New Site button	click, <Tab>, mnemonic (N) or accelerator (N)	spawns New Site dialog box

Table 7. Customer Manager CAR Diagram

Control Area	User behavior	System response
File>Existing Site button	click, <Tab>, mnemonic (E), or accelerator (E)	spawns Existing Site dialog box. Specific design of the dialog box will have to be determined.
File>Save As Site button	click, <Tab>, mnemonic (S), or accelerator (S)	saves customer site profile information; if site is not specified, spawns Customer Site Error dialog box. Specific design of the dialog box will have to be determined.
File>Exit button	click, <Tab>, mnemonic (X), or accelerator (X)	spawns a confirmatory dialog box ("Do you really want to exit this application?" OK Cancel); if user selects OK, closes the application; if user selects Cancel, removes dialog box.
Edit cascade button	click, <Tab>	spawns a pull-down menu displayed below Edit cascade button contents TBD
View cascade button	click, <Tab>, mnemonic (V), or accelerator (V)	spawns a pull-down menu displayed below View cascade button
View>Profile Page 1 (or 2)	click, <Tab>, mnemonic (P), or accelerator (P)	displays the Profile pane (1 or 2)
View>Summary Page	click, <Tab>, mnemonic (U), or accelerator (U)	displays the Summary pane
Application cascade button	click, <Tab>, mnemonic (A), or accelerator (A)	spawns a pull-down menu displayed below Application cascade button
Application>Recommendation button	click, <Tab>, mnemonic (R), or accelerator (R)	spawns Recommendation window if site is specified
Application>Product Manager button	click, <Tab>, mnemonic (M), or accelerator (M)	spawns Product Manager window
Help cascade button	click, <Tab>	spawns Help dialog box contents TBD
Control Area II: ICONIC BUTTON BAR		
Profile 1 or 2 icon	click, <Tab>	displays Profile 1 or 2 pane
Summary icon	click, <Tab>	displays Summary pane
Recommendation icon	click, <Tab>	spawns Recommendation window

Table 7. Customer Manager CAR Diagram

Control Area	User behavior	System response
Product Editor icon	click, <Tab>	spawns Product Editor window
Help Off/On icon	click, <Tab>	turns multi-line help area off or on; default is on
Help icon	click, <Tab>	contents TBD
Control Area III: CUSTOMER BANNER		
read only text field (80 characters)	none	displays customer name in all caps
Control Area IV: HELP TEXT DISPLAY FIELD		
Read only text field	none	displays help message related to location of mouse pointer

Table 7. Customer Manager CAR Diagram

6.4 Customer Site Profile Pane Layout and Behavior

There are two profile panes; they display 56 profile items in nine categories for H&PS and 78 profile items in 16 categories for SBG. The profile panes display in the Sales Helper Window one at a time; the SSC controls the display through iconic push buttons or menu bar. The organization of the profile categories and items is determined by the following considerations:

- the natural flow of the customer contacts
- the frequency in which profile categories and items are visited
- the number of follow-on questions that will fall within the same page.

The H&PS and SBG SMEs have determined flow and frequency.

There are two types of profile items on the Sales Helper. Table 8. describes their widget representations.

Profile Item Type	Widget Representation
Yes/No/Unknown	Custom Tri-State Toggle Buttons
Mutually-Exclusive	Option Menu

Table 8. Widget Representations of Profile Items

The categories and the profile items are organized into three columns with the following rules:

- No category can span more than one column without having category names on both columns. If the length of a category extends beyond a pre-determined size, it will either be placed in another column or a scrolling mechanism will be provided to the user. The choice of specific mechanism will have to be determined.

- A category cannot span over two panes.
- Text labels associated with each profile item cannot exceed 22 characters in length. Extra characters will be truncated in the display.
- Items in an option menu cannot exceed 15 characters in length. Extra characters will be truncated in the display.
- The text labels of the profile categories will be left justified to the corresponding profile category pane and be placed above the border of the pane.
- The text labels of the profile items will be left-justified and indented five characters from the left border.
- All text labels will display in mixed case.

The change of background color (from original to green and from green to original) of follow-on profile items is effected by the following rules:

- If the states of the follow-on questions are known, the background will remain as the original color.
- If the states of the follow-on questions are unknown, the background will change to green. Once the states are known, the background will change back to the original background.
- If the follow-on questions are on a different page, the background of the Next (or Previous) page push button at the lower right corner of the window will change to green. It will change back to its original color after the other page is visited.

SSCs use the menu bar View cascade menu or the available Profile page icon button(s) to access these panes from the Sales Helper window. The SSC enters information by pressing the left mouse button at the appropriate control.

6.5 Sales Helper SBG and HPS Business Function

The SSC uses the SBG and HPS Sales Helper: Profile panes to gather, retrieve, and edit customer site profile information. The different panes of profile information, specific to the market unit of the customer site, are displayed in the Sales Helper window through user control.

In the initial phase of CCOE Sales Helper, the profile items may have to be stored locally by the application because of technology limitations of CSDB. The long-term goal is to have the screens dynamically built from the information in the knowledge base and the CSDBs. In this way, the user interface does not have any knowledge of the profile items, and their interrelationships—such as follow-on profile items and changes of profile items—can be done without changing the code of the user interface. For the sake of clarity and completeness, the follow-on questions are included in this specification to facilitate prototype construction and quick reference.

The Sales Helper Profile panes support the following business functions:

- displaying existing customer site profile information
- entering new customer site profile information
- updating existing customer site profile information
- resetting customer site profile information.

The customer site profile information collected by the market units in Mass Markets differs significantly. To support both channels, four different panes represent the information requirements for SBG and HPS. The application determines which panes to display, based on the customer's primary market unit assignment. Section 6.6 and its subsections describe the SBG Profile panes. Section 6.7 and its subsections describe the HPS Profile panes.

6.6 SBG Profile Panes

The SBG Sales Helper Profile 1 and 2 panes (Figure 18. and Figure 19.) allow the SSC to gather information about the business, layout, concerns, usage, equipment, and other customer needs.

CCOE: SBU Sales Helper Profile 1 (Test 9)

File Edit View Application Help

Customer

Business Info		Yes		No		Concerns		Yes		No		Identify Calls		Yes		No			
New / Startup		☐	☐		☐	Missed Calls		☐	☐		☐	Screen		☐	☐		☐		
Moving		☐	☐		☐	Busy Line		☐	☐		☐	Track Calls In		☐	☐		☐		
Growth	Unknown					Phone Tag		☐	☐		☐	Control Calls		☐	☐		☐		
Multi-Business		☐	☐		☐	Lost Messages		☐	☐		☐	Restrict In		☐	☐		☐		
Multi-Users		☐	☐		☐	Priority Calls		☐	☐		☐	Restrict Out		☐	☐		☐		
Multi-Line Line		☐	☐		☐	Privacy		☐	☐		☐	Urgent		☐	☐		☐		
Work Home Occasionally		☐	☐		☐	Annoyance Calls		☐	☐		☐	Ans Methods		☐	☐		☐		
Call Motivated by				Easy Calling								Self				☐			
Media	Unknown					Easy to Reach		☐	☐		☐	Others		☐	☐		☐		
Promotion	Unknown					Easy Number		☐	☐		☐	After Hours		☐	☐		☐		
Layout												Out of Office							
# Staff	Unknown					Save Messages		☐	☐		☐	Work Out of Office		☐	☐		☐		
Depts / Offices		☐	☐		☐	Freq Dialed #s		☐	☐		☐	Voice		☐	☐		☐		
Work Areas		☐	☐		☐	Conference		☐	☐		☐	Message Notify		☐	☐		☐		
Multi-Premises		☐	☐		☐	Info Only		☐	☐		☐	On Call		☐	☐		☐		
												Last Update 5-25-95							
																Next			

Figure 18. SBU Sales Helper Profile 1

In / Out Usage		Equipment		Visibility	
	Yes No		Yes No		Yes No
Local In	Unknown	# Phone Sets	Unknown	Local	☐ ☐
Local Out	Unknown	Cordless	☐ ☐	Key Cities	☐ ☐
IntraLATA In	Unknown	Cellular	Unknown	State	☐ ☐
IntraLATA Out	Unknown	Pager	Unknown	National	☐ ☐
InterLATA In	Unknown	Fax	Unknown	Advertising	
InterLATA Out	Unknown	Modem	Unknown	Phone Book	☐ ☐
Data In	Unknown	Credit Card Mch	Unknown	Multimedia	☐ ☐
Data Out	Unknown	Ans Mch	Unknown	Customer	
		Ans Svc	☐ ☐	New	☐ ☐

Usage		Image	
	Yes No		Yes No
Seasonal	☐ ☐	Big	☐ ☐
Extended Absence	☐ ☐	Customized	☐ ☐

Customer	
	Yes No
Repeat	☐ ☐
Walk-ins	☐ ☐
Referrals	☐ ☐

Last Update: 5-25-95
Previous

Figure 19. SBG Sales Helper Profile 2

The behavior of widgets in this window is described in Section 6.4 . The multi-line help has been turned off in Figure 18, and Figure 19.

6.6.1 Sales Helper SBG Profile Pane Edits

No edits have been identified at this time.

6.6.2 Sales Helper SBG Profile Pane Field Descriptions

The field descriptions of individual profile items and categories are included in Appendix A. The field descriptions of other controls are in section 6.4 .

6.6.3 Sales Helper SBG Profile CAR Diagram

This section captures interface behavior as specified for each control on the Sales Helper SBG Profile window. The following tables specify each control on the Sales Helper SBG Profile window and panes, the allowable user

actions, and the corresponding system response associated with each of these controls. Table 9. represents the Profile 1 pane, and Table 10. the Profile 2 pane.

For user behavior, the following notation will be used:

Move = user moves mouse to area of widget, field help will be displayed in the multi-line help area

click = user clicks on item (button) using mouse button 1.

<Tab> = user tabs to item (button) and then <Return>.

Mnemonic (key) = user activates mnemonic by <Alt> and the key indicated in parentheses.

Accelerator (key) = user activates accelerator by <Control> and the key indicated in parentheses.

Control	User behavior	System response
Control Area I: WINDOW MENU BAR		
File cascade button	TBD	TBD
Edit cascade button	TBD	TBD
View cascade button	TBD	TBD
Application cascade button	TBD	TBD
Help cascade button	TBD	TBD
Control Area II: ICONIC PUSH BUTTON BAR		
Profile Page icon	move click	displays field help displays next profile page
Summary icon	move click	displays field help displays profile summary page
Recommendation icon	move click	displays field help invokes and displays recommendation window
Product Manager icon	move click	displays field help invokes and displays product manager
Help Off/On icon	move click	displays field help turns multi-line help area off or on; default is on
Help icon	move click	displays field help TBD
Control Area III: CUSTOMER BANNER		
read only text field (80 characters)	none	displays customer name in all caps
Control Area IV: BUSINESS INFO PANE		

Table 9. SBG Sales Helper Profile 1 CAR Diagram

Control	User behavior	System response
Yes No text label	none—system generated	none
Business Info category text label and pane	move	displays field help
New/Start-up text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Track Calls In (Identify Calls)
Moving text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No Follow-on.
Growth text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No Follow-on.
Multi-Business text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Screen (Identify Calls) •Others (Ans Method)
Multi-Users text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Multi-Use Line (Business Info)

Table 9. SBG Sales Helper Profile 1 CAR Diagram

Control	User behavior	System response
Multi-Use Line text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Multi-Users (Business Info)
Work Home Occasionally text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Message Notify (Out of Office) •Transfer (Easy Calling)
Control Area V: CALL MOTIVATED BY PANE		
Call Motivated by category text label and pane	move	displays field help
Media text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No Follow-on.
Promotion text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No Follow-on.
Control Area VI: LAYOUT PANE		
Layout category text label and pane	move	displays field help

Table 9. SBG Sales Helper Profile 1 CAR Diagram

Control	User behavior	System response
# Staff text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (5-9,10+): •Busy Line (Concerns) •Waiting for Line (Concerns) •Phone Tag (Concerns) •Lost Messages (Concerns) •Privacy (Concerns) •Depts/Offices (Layout)
Depts/Offices text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Multi-Use Line (Business Info) •Multi-Users (Business Info)
Work Areas text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Self (Ans Methods) •Others (Ans Methods)
Multi-Premises text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Work Out of Office (Out of Office) •Pager (Equipment) •Others (Ans Methods)
Control Area VII: CONCERNS PANE		
Yes No text label	none—system generated	none
Concerns category text label and pane	move	displays field help.

Table 9. SBG Sales Helper Profile I CAR Diagram

Control	User behavior	System response
Missed Calls text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Multi-Use Line (Business Info)
Busy Line text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Multi-Users (Business Info)
Phone Tag text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Ans Mch (Equipment)
Lost Messages text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Ans Mch (Equipment) •Save Messages (Easy Calling)
Priority Callers text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Easy Number (Easy Calling) •Screen (Identify Calls In)

Table 9. SBC Sales Helper Profile 1 CAR Diagram

Control	User behavior	System response
Privacy text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Multi-Users (Business Info) •Work Areas (Layout)
Annoyance Calls text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Screen (Identify Calls In)
Control Area VII: EASY CALLING PANE		
Easy Calling category text label and pane	move	displays field help
Easy to Reach text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions.
Easy Number text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions.
Save Messages text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions.

Table 9. SBG Sales Helper Profile 1 CAR Diagram

Control	User behavior	System response
Freq Dialed #s text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Repeat (Customers)
Conference text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): No follow-on questions
Info Only text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): No follow-on questions
Transfer text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Others (Ans Methods) •Priority Callers (Concerns)
Control Area VIII: IDENTIFY CALLS PANE		
Yes No text label	none—system generated	none
Identify Calls category text label and pane	move	displays field help
Screen text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •After Hours (Ans Methods) •Priority Callers (Concerns)

Table 9. SBG Sales Helper Profile 1 CAR Diagram

Control	User behavior	System response
Track Calls In text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Multi-Users (Business Info)
Control Area IX: CONTROL CALLS PANE		
Control Calls category text label and pane	move	displays Field help
Restrict In text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Annoyance Calls (Concerns) •Screen (Identify Calls In)
Restrict Out text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions
Urgent text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •On Call (Out of Office) •Others (Ans Methods) •Message Notify (Out of Office)
Control Area X: ANS METHODS PANE		
Ans Methods category text label and pane	move	displays field help.

Table 9. SBG Sales Helper Profile 1 CAR Diagram

Control	User behavior	System response
Self text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Work Out of Office (Out of Office) •Work Areas (Layout)
Others text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Work Out of Office (Out of Office) •Work Areas (Layout)
After Hours text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions
Control Area XI: OUT OF OFFICE PANE		
Out of Office category text label and pane	move	displays field help
Work Out of Office text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Others (Ans Method) •Save Messages (Easy Calling)

Table 9. SBG Sales Helper Profile 1 CAR Diagram

Control	User behavior	System response
Voice text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Others (Ans Method) •Cellular (Equipment)
Message Notify text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Others (Ans Method) •Pager (Equipment)
On Call text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Others (Ans Method) •Pager (Equipment) •Cellular (Equipment)
Control Area XII: MULTI-LINE HELP AREA		
multi-line help	none—system generated	displays field help when the mouse cursor is over the corresponding widget or text label
Control Area XIII: LAST UPDATE DISPLAY		
Last Update date	none—system generated	displays the last update date of profile items
Control Area XIV: NEXT PAGE PUSH BUTTON		
Next Page Push Button	click, <Tab>	Change the current profile pane to the next page If follow-on questions appear on a different page, the background of this button changes to green. It will change back to the original color after the SSC visits the page.

Table 9. SBG Sales Helper Profile 1 CAR Diagram

Control	User behavior	System response
Control Area I: WINDOW MENU BAR		
File cascade button	TBD	TBD
Edit cascade button	TBD	TBD
View cascade button	TBD	TBD
Application cascade button	TBD	TBD
Help cascade button	TBD	TBD
Control Area II: ICONIC PUSH BUTTON BAR		
Profile Page icon	move click	displays field help displays previous profile page
Summary icon	move click	displays field help displays profile summary page
Recommendation icon	move click	displays field help invokes and displays recommendation window
Product Manager icon	move click	displays field help invokes and displays product manager
Help Off/On icon	move click	displays field help turns multi-line help area off or on; default is on
Help icon	move click	displays field help TBD
Control Area III: CUSTOMER BANNER		
read only text field (80 characters)	none	displays customer name in all caps
Control Area IV: IN/OUT USAGE PANE		
Yes No text label	none—system generated	none
In/Out Usage category text label and pane	move	displays field help

Table 10. SBG Sales Helper: Profile 2 CAR Diagram

Control	User behavior	System response
Local In text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (High): •Waiting for Line (Concerns) •Busy Line (Concerns) •Privacy (Concerns) •Multi-Use Line (Business Info)
Local Out text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Waiting for Line (Concerns) •Busy Line (Concerns) •Privacy (Concerns) •Multi-Use Line (Business Info)
IntraLATA In text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (High): •Busy Line (Concerns) •Multi-Use Line (Business Info)
IntraLATA Out text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (High): •Busy Line (Concerns) •Multi-Use Line (Business Info)

Table 10. SBG Sales Helper: Profile 2 CAR Diagram

Control	User behavior	System response
InterLATA In text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions
InterLATA Out text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions
Data In text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (High): •Modem (Equipment) •Fax (Equipment) •Multi-Use Line (Business Info)
Data Out text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (High): •Modem (Equipment) •Fax (Equipment) •Multi-Use Line (Business Info)
Control Area V: USAGE PANE		
Usage category text label and pane	move	displays field help
Seasonal text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions

Table 10. SBG Sales Helper: Profile 2 CAR Diagram

Control	User behavior	System response
Extended Absence text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Seasonal (Usage) •Work Out of Office (Out of Office)
Control Area VI: EQUIPMENT PANE		
Yes No text label	none—system generated	none
Equipment category text label and pane	move	displays field help
# Phone Sets text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions
Cordless text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Depts/Offices (Layout) •Work Areas (Layout)
Cellular text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions

Table 10. SBG Sales Helper: Profile 2 CAR Diagram

Control	User behavior	System response
Pager text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Own/Buy): •On Call (Out of Office) •Urgent (Control Calls) •Work Out of Office (Out of Office)
Fax text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Own/Buy): •Multi-Use Line (Business Info) •Data In (In/Out Usage) •Data Out (In/Out Usage)
Modem text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Own/Buy): •Multi-Use Line (Business Info) •Data In (In/Out Usage) •Data Out (In/Out Usage)
Credit Card Mch text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Own/Buy): •Multi-Use Line (Business Info) •Data In (In/Out Usage) •Data Out (In/Out Usage)

Table 10. SBG Sales Helper: Profile 2 CAR Diagram

Control	User behavior	System response
Ans Mch text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Own/Buy): •Privacy (Concerns) (Problem): •Lost Messages (Concerns) (No) •Lost Messages (Concerns) •Save Messages (Easy Calling)
Ans Svc text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions
Control Area VII: IMAGE PANE		
Image category text label and pane	move	displays field help
Big text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions
Customized text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions
Control Area VIII: VISIBILITY PANE		
Yes No text label	none—system generated	none
Visibility category text label and pane	move	displays field help

Table 10. SBG Sales Helper: Profile 2 CAR Diagram

Control	User behavior	System response
Local text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions
Key Cities text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions
State text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions
National text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions
Control Area IX: ADVERTISING PANE		
Yes No text label	none—system generated	none
Advertising category text label and pane	move	displays field help
Phone Book text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Repeat (Customer)
MultiMedia text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions

Table 10. SBG Sales Helper: Profile 2 CAR Diagram

Control	User behavior	System response
Control Area X: CUSTOMER PANE		
Yes No text label	none—system generated	none
Customer category text label and pane	move	displays field help
New text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Phone Book (Advertising)
Repeat text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Phone Book (Advertising)
Walk-Ins text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Phone Book (Advertising)
Referrals text label and tri-state toggle button	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Phone Book (Advertising)
Control Area XI: MULTI-LINE HELP AREA		
multi-line help	none—system generated	displays field help when the mouse cursor is over the corresponding widget or text label
Control Area XII: LAST UPDATE DISPLAY		
Last Update date	none—system generated	displays the last update date of profile items

Table 10. SBG Sales Helper: Profile 2 CAR Diagram

Control	User behavior	System response
Control Area XIII: PREVIOUS PAGE PUSH BUTTON		
Previous Page push button	click, <Tab>	change the current profile pane to the previous page If follow-on questions appear on a different page, the background of the button changes to green. It will change back to the original color after the SSC visits the page.

Table 10. SBG Sales Helper: Profile 2 CAR Diagram

6.7 HPS Profile Panes

The HPS Sales Helper Profile 1 and 2 panes (Figure 18. and Figure 19.) allow the SSC to gather information about the home service, layout, and equipment; and the customer's life style, concerns, and other needs.

The icons used in these windows are placeholders. Additional design and art work will have to be done.

The multi-line help has been turned on in Figure 18. and Figure 19. The behavior of the widgets in this window is described in section 6.4 .

CCOE - HPS Sales Helper - Profile 1

File Edit View Application Help

Customer

Service Status		Equipment	
	Yes No		Yes No
New Service	☐ ☐	# Phone Sets	Unknown
Working	☐ ☐	Ant Mch / Sys	☐ ☐
Total Deactivate	Unknown	Pager	☐ ☐

Layout

Home Type	One Level
Detached Locs	Multi Level
# Jacks	Apartment
Alarm	Trailer
	Other
	Unknown

Call Motivated by

Media	Unknown
Promotion	Unknown

Usage

In Local	Unknown
Out Local	Unknown
Toll In LATA	Unknown

Structure of home including levels, living space, and work areas

Last Update
5-25-1995

Next

Figure 20. HPS Sales Helper Profile Pane 1

Phone Users		Life Style		Customer Needs	
	Yes No		Yes No		Yes No
# Users	4+	Commute to Work	☐ ☐	Easy to Reach	☐ ☐
Spouse	☐ ☐	Home Office	☐ ☐	Can't Miss Calls	☐ ☐
Roommates	☐ ☐	Travel Frequently	☐ ☐	Privacy within Home	☐ ☐
# Children	1 - 2	Out and About	☐ ☐	Home Members Unique ID	☐ ☐
Child under 5 Yrs	☐ ☐	Extended Absence	☐ ☐	Save Messages	☐ ☐
Child 5 - 10 Yrs	☐ ☐			Screen Calls	☐ ☐
Child 11 - 18 Yrs	☐ ☐	Concerns		Simultaneous Usage	☐ ☐
Family in Other Locs	☐ ☐	Busy Line	☐ ☐	Restrict in Local	☐ ☐
Additional Users	Unknown	Waiting for Line	☐ ☐	Restrict in Toll	☐ ☐
Special Needs	☐ ☐	Annoyance Calls	☐ ☐	Restrict Out Toll	☐ ☐
Budget	Unknown	Lost Messages	☐ ☐	Conferences	☐ ☐
		Phone Tag	☐ ☐	Frequently Dialed #s	☐ ☐
		Mixed Calls	☐ ☐	Easy Callback	☐ ☐
		Misdialed #s Out	☐ ☐	Easy Number	☐ ☐

Additional household users such as parents, extended family, etc.

Last Update
5-25-1993

Previous

Figure 21. HPS Sales Helper Profile Pane 2

These figures show a typical consultative session with various follow-on questions. Beginning in Figure 20., the SSC is about to choose an option from the profile item Home Type. The background of both Home Type and # Jacks has turned green to reflect their follow-on status from the Service Status category, where the New Service item is set to the Yes state. The Equipment category Modem/Fax item is also set to Yes, which has identified some follow-on questions on Profile 2. The Next push button in Profile 1 has turned green to reflect those follow-on questions. Figure 21. shows the follow-on questions with the background of the corresponding widgets in green. The follow-on items will change back to their original background color if their states become known.

6.7.1 Sales Helper HPS Profile Pane Edits

No edits have been identified at this time.

6.7.2 Sales Helper HPS Profile Pane Field Descriptions

The field descriptions of individual profile items and categories are included in Appendix A. The field descriptions of other controls are in Section 6.4.

6.7.3 Sales Helper HPS Profile CAR Diagram

This section captures interface behavior as specified for each control on the Sales Helper HPS Profile window. The following tables specify each control on the Sales Helper HPS Profile window and panes, the allowable user actions, and the corresponding system response associated with each of these controls. Table 11. represents the Profile 1 pane, and Table 12. the Profile 2 pane.

For user behavior, the following notation will be used:

Move = user moves mouse to area of widget

click = user clicks on item (button) using mouse button 1.

<Tab> = user tabs to item (button) and then <Return>.

Mnemonic (key) = user activates mnemonic by <Alt> and the key indicated in parentheses.

Accelerator (key) = user activates accelerator by <Control> and the key indicated in parentheses.

Control	User behavior	System response
Control Area I: WINDOW MENU BAR		
File cascade button	TBD	TBD
Edit cascade button	TBD	TBD
View cascade button	TBD	TBD
Application cascade button	TBD	TBD
Help cascade button	TBD	TBD
Control Area II: ICON BAR		
Profile Page icon	move click	displays field help displays next Profile page
Summary icon	move click	displays field help displays profile Summary
Recommendation icon	move click	displays field help invokes Recommendation window
Product Manager icon	move click	displays field help invokes Product Manager window

Table 11. Sales Helper HPS Profile 1 CAR Diagram

Control	User behavior	System response
Help Off/On icon	move click	displays field help turns multi-line help area off or on; default is on
Help icon	move click	displays field help TBD
Control Area III: CUSTOMER BANNER		
read only text field (80 characters)	none	displays customer name in all caps
Control Area IV: SERVICE STATUS PANE		
Yes No text label	none—system generated	none
Service Status category text label and pane	move	displays field help
New Service text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Home Type (Layout) •# of Jacks (Layout)
Moving text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Easy to Reach (Customer Needs)
Total Deactivate text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions.
Layout category text label and pane	move	displays field help

Table 11. Sales Helper HPS Profile 1 CAR Diagram

Control	User behavior	System response
Home Type text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (all options): •# Jacks (Layout)
Detached Locs text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Easy to Reach (Customer Needs) •Lost Messages (Concerns)
# Jacks text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (all options): •Home Type (Layout)
Alarm text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions.
Control Area V: CALL MOTIVATED BY PANE		
Call Motivated by category text label and pane	move	displays field help
Media text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions.

Table 11. Sales Helper HPS Profile 1 CAR Diagram

Control	User behavior	System response
Promotion text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions.
Control Area VI: EQUIPMENT PANE		
Yes No text label	none—system generated	none
Equipment category text label and pane	move	displays field help
# Phone Sets text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (all options): •Simultaneous Usage (Customer Needs)
Ans Mch/Svc text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Miss Calls when away (Concerns) •Busy Line (Concerns)
Pager text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Out and About (Life Style)

Table 11. Sales Helper HPS Profile 1 CAR Diagram

Control	User behavior	System response
Cellular/Mobile text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Out and About (Life Style)
Modem/Fax text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Simultaneous Usage (Customer Needs)
Control Area VII: USAGE PANE		
Usage category text label and pane	move	displays field help
In Local text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (11-20 Mins): •Busy Line (Concerns) (21+ Mins): •Busy Line (Concerns) •Waiting for Line (Concerns)
Out Local text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (11-20 Mins): •Waiting for Line (Concerns) (21+ Mins): •Busy Line (Concerns) •Waiting for Line (Concerns)

Table 11. Sales Helper HPS Profile 1 CAR Diagram

Control	User behavior	System response
Toll in LATA text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Medium): •Busy Line (Concerns) (High): •Busy Line (Concerns) •Waiting for Line (Concerns)
Control Area VIII: MULTI-LINE HELP AREA		
multi-line help	none—system generated	displays field help when the mouse cursor is over the corresponding widget or text label
Control Area IX: LAST UPDATE DISPLAY		
Last Update date	none—system generated	displays the last update date of profile items
Control Area X: NEXT PAGE PUSH BUTTON		
Next Page push button	click, <Tab>	Change the current profile pane to the next page If follow-on questions appear on a different page other than the current page, the background of the button will be changed to green. It will change back to the original color after the user visits the page by clicking on it.

Table 11. Sales Helper HPS Profile 1 CAR Diagram

Control	User behavior	System response
Control Area I: WINDOW MENU BAR		
File cascade button	TBD	TBD
Edit cascade button	TBD	TBD
View cascade button	TBD	TBD
Application cascade button	TBD	TBD

Table 12. Sales Helper HPS Profile 2 CAR Diagram

Control	User behavior	System response
Help cascade button	TBD	TBD
Control Area II: ICON BAR		
Profile Page icon	move click	displays field help displays previous profile page
Summary icon	move click	displays field help displays profile summary page
Recommendation icon	move click	displays field help invokes and displays recommendation window
Product Manager icon	move click	displays field help invokes and displays product manager
Help Off/On icon	move click	displays field help turns multi-line help area off or on; default is on
Help icon	move click	TBD
Control Area III: CUSTOMER BANNER		
read only text field (80 characters)	none	displays customer name in all caps
Control Area IV: PHONE USERS PANE		
Yes No text label	none—system generated	none
Phone Users category text label and pane	move	displays field help
# Users text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (2, 3-4, 5+): •Busy Line (Concerns) •Waiting for Line (Concerns)

Table 12. Sales Helper HPS Profile 2 CAR Diagram

Control	User behavior	System response
Spouse text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Home Member Unique ID (Customer Needs)
Roommates text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Can't Miss Calls (Customer Needs) •Busy Line (Concerns) •Waiting for Line (Concerns) •Save Messages (Customer Needs) •Lost Messages (Concerns)
# Children text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (3+): •Busy Line (Concerns) •Waiting for Line (Concerns)
Child under 5 Yrs. text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Freq Dialed #s (Customer Needs)

Table 12. Sales Helper HPS Profile 2 CAR Diagram

Control	User behavior	System response
Child 5 - 10 Yrs text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Freq Dialed #s (Customer Needs) •Restrict Out Toll (Customer Needs)
Child 11 - 18 Yrs text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Home Member Unique ID (Customer Needs) •Busy Line (Concerns) •Waiting for Line (Concerns) •Conference (Customer Needs) •Restrict Out Toll (Customer Needs)
Family in Other Locs text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Restrict In Toll (Customer Needs) •Conference (Customer Needs) •Freq Dialed #s (Customer Needs)

Table 12. Sales Helper HPS Profile 2 CAR Diagram

Control	User behavior	System response
Additional Users text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (all options): •Restrict Out Toll (Customer Needs) •Privacy within Home (Customer Needs) •Busy Line (Concerns) •Waiting for Line (Concerns) •Home Members Unique ID (Customer Needs)
Special Needs text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions
Budget text label and option menu	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (<\$20 Month): •Restrict In Toll (Customer Needs) •Restrict Out Toll (Customer Needs) •Toll in LATA (Usage) •Local Out (Usage)
Control Area V: LIFE STYLE PANE		
Yes No text label	none—system generated	none
Life Style category text label and pane	move	displays field help

Table 12. Sales Helper HPS Profile 2 CAR Diagram

Control	User behavior	System response
Commute to Work text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): • Can't Miss Calls (Customer Needs) • Miss Calls When Away (Concerns)
Home Office text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): • Easy Number (Customer Needs) • Conference (Customer Needs)
Travels Frequently text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on Questions
Out and About text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): • Can't Miss Calls (Customer Needs)
Extended Absence text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): • Easy to Reach (Customer Needs)
Control Area VI: CONCERNS PANE		

Table 12. Sales Helper HPS Profile 2 CAR Diagram

Control	User behavior	System response
Concerns category text label and pane	move	displays field help
Busy Line text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Save Messages (Customer Needs)
Waiting for Line text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Save Messages (Customer Needs)
Annoyance Calls text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Screen Calls (Customer Needs)
Lost Messages text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Ans Mch/Svc (Equipment)
Phone Tag text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Home Office (Life Style) •Easy Callback (Customer Needs)

Table 12. Sales Helper HPS Profile 2 CAR Diagram

Control	User behavior	System response
Missed Calls text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): • Commute to Work (Life Style) • Travels Frequently (Life Style)
Misdialed #s Out text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): • # of Children (Phone Users)
Control Area VII: CUSTOMER NEEDS PANE		
Yes No text label	none—system generated	none
Customer Needs category text label and pane	move	displays field help
Easy to Reach text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): • Home Members Unique ID (Customer Needs) • Travels Frequently (Life Style)
Can't Miss Calls text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): • Miss Calls (Concerns) • Out and About (Life Style) • Travels Frequently (Life Style)

Table 12. Sales Helper HPS Profile 2 CAR Diagram

Control	User behavior	System response
Privacy within Home text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Home Office (Life Style) •# Users (Phone Users)
Home Members Unique ID text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Easy Number (Customer Needs)
Save Messages text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Busy Line (Concerns)
Screen Calls text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. No follow-on questions
Simultaneous Usage text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Modem/Fax (Equipment) •Busy Line (Concerns) •Waiting for Line (Concerns)

Table 12. Sales Helper HPS Profile 2 CAR Diagram

Control	User behavior	System response
Restrict In Toll text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Toll in LATA (Usage)
Restrict Out Toll text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Toll in LATA (Usage)
Conference text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Home Office (Life Style)
Frequently Dialed #s text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Local Out (Usage)
Easy Callback text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Annoyance Calls (Concerns)

Table 12. Sales Helper HPS Profile 2 CAR Diagram

Control	User behavior	System response
Easy Number text label and tri-state toggle buttons	move click or <Tab>	displays field help updates to the user choice Turns off follow-on status of item if it is on and the state of the profile item is known. Follow-on Profile Items (Yes): •Home Members Unique ID (Customer Needs)
Control Area VIII: MULTI-LINE HELP AREA		
multi-line help	none—system generated	displays field help when the mouse cursor is over the corresponding widget or text label
Control Area IX: LAST UPDATE DISPLAY		
Last Update date	none—system generated	displays the last update date of profile items
Control Area X: PREVIOUS PAGE PUSH BUTTON		
Previous Page push button	click, <Tab>	changes the current profile pane to the previous page If follow-on questions appear on a different page other than the current page, the background of the button will be changed to green. It will change back to the original color after the user visits the page by clicking on it.

Table 12. Sales Helper HPS Profile 2 CAR Diagram

6.8 Profile Summary Pane

When the SSC invokes the Sales Helper for customers with an existing site, the Sales Helper Profile Summary pane displays first. The Profile Summary provides a quick way of seeing the profile characteristics of the customer site. Its function is to help the SSC establish rapport with the customer. The SSC can use iconic push buttons or the menu bar to move back and forth between the Profile Summary pane and the Profile 1 and 2 panes, the Recommendation window, or the Product Manager window.

The Profile Summary pane is a read-only pane with no data entry fields. It presents the known values for the profile items within categories in the same order that they appear in the Sales Helper Profile 1 and 2 panes. The display is organized into three columns, reading from left to right and top to bottom. If the information exceeds the predetermined size of the Sales Helper Profile Summary pane, a vertical scroll bar will be provided.

The item display is constructed dynamically from the current known values of the profile panes, and the value of each item is enclosed by parentheses after the text label of the profile item.

The Sales Helper: Profile Summary window in Figure 22. contains a Profile Summary pane for an H&PS customer site. The categories and items will differ for a SBG customer site.

CCOE: H&PS Sales Helper: Profile Summary

File Edit View Application Help

Customer

Service Status
Moving (Yes)

Layout
Home Type (Multi-Level)
Jacks (3)

Equipment
Phone Sets (3)
Modem / Fax (Yes)

Usage
In Local (21+ min day)
Toll in LATA (Low)

Phone Users
Users (4)
Spouse (Yes)
of Children (2)
Child 5-10 Yrs (Yes)
Child 11-18 Yrs (Yes)

Life Style
Commutate to Work (No)
Home Office (Yes)

Concerns
Waiting for Lines (Yes)

Customer Needs
Separate ID (Yes)
Can't Miss Calls (No)

Last Update 5-25-1995

Figure 22. CCOE Sales Helper Profile Summary Pane

6.8.1 Sales Helper Profile Summary Pane Field Descriptions

The fields are dynamically constructed from the profile item entries (known values) and are read-only. For a detailed descriptions of the profile items and their meanings, see Appendix A in this report.

6.8.2 Sales Helper Profile Summary Pane Edits

There are no pane edits identified.

6.8.3 Sales Helper Profile Summary Pane CAR Diagram

This section captures interface behavior as specified for each control on the Sales Helper Profile Summary window. The following table specifies each control on the Sales Helper Profile Summary window, the allowable user actions, and the corresponding system response associated with each of these controls.

For user behavior, the following notation will be used:

click = user clicks on item (button) using mouse button 1

<Tab> = user tabs to item (button) and then <Return>

mnemonic (key) = user activates mnemonic by <Alt> and the key indicated in parentheses

accelerator (key) = user activates accelerator by <Control> and the key indicated in parentheses

Control	User Behavior	System Behavior
Control Area I: WINDOW MENU BAR		
File cascade button	TBD	TBD
Edit cascade button	TBD	TBD
View cascade button	TBD	TBD
Application cascade button	TBD	TBD
Help cascade button	TBD	TBD
Control Area II: ICONIC PUSH BUTTON BAR		
Profile Page 1 icon	move click	displays field help displays profile screen 1
Profile Page 2 icon	move click	displays field help displays profile screen 2
Recommendation icon	move click	displays field help invokes Recommendation window
Product Manager icon	move click	displays field help invokes Product Manager window
Help Off/On icon	move click	displays push button field help turns multi-line help area off or on; default is on

Table 13. Sales Helper Profile Summary Pane CAR Diagram

Control	User Behavior	System Behavior
Help icon	move click	displays field help TBD
Control Area III: CUSTOMER BANNER		
read only text field (80 characters)	none	displays customer name in all caps
Control Area IV: PROFILE SUMMARY PANE		
Profile Summary pane	none—system generated	none
Control Area V: MULTI-LINE HELP AREA		
multi-line help	none—system generated	displays field help when the mouse cursor is over the corresponding widget or text label
Control Area VI: LAST UPDATE DISPLAY		
Last Update date	none—system generated	displays the last update date of profile items

Table 13. Sales Helper Profile Summary Pane CAR Diagram

6.9 Recommendation Window Specifications

Based on the customer site profile information gathered in the interview with the customer, the Recommendation Window (Figure 23.) displays different possible sets of solutions recommended by the knowledge-based system. It allows the SSC to present the proposed solution to satisfy customer needs and concerns.

The solutions recommended by the system fall into three types of mapping to U S WEST products:

- The solution corresponds to one U S WEST product.
It will be represented by a push button with the solution/product name on it. The catalog rate for recurring charge (monthly) and non-recurring charge (one-time) will be enclosed in square brackets after the product name. Clicking on the push button will add the product to the list in the Proposed Product List window. Double-clicking on the push button will invoke the Product Manager window with the corresponding product screen.
- The solution corresponds to a generic family of U S WEST products having different flavors, such as call forwarding.
It will be represented by a push button followed by ellipses. Clicking on the push button will invoke a dialog box containing the different flavors of the solution with the catalog rate for each of the flavors. The SSC can choose one or more flavors through check boxes. After the choices are

made, the SSC can either click OK to confirm the choices and put them into the proposed list or cancel the choices. Double-clicking on the push button will invoke the Product Manager with the corresponding screen.

- The solution corresponds to a combination of different solutions, such as call waiting and three-way calling. Each of the solutions may fall into either of the first two types. Each of the solutions will be represented by a push button with its name followed by ellipses on the same line. Clicking on the push button will invoke a dialog box containing the catalog rates of the constituent solution. If the constituent solution corresponds to a generic family of products, it will be handled in the same way as described above. Double-clicking on the push button will invoke the Product Manager with the corresponding screen.

The possible solutions are organized by the capabilities that the solutions provide. Each capability displays a maximum of three possible associated solutions, arranged in decreasing order of weight as determined by the knowledge engine. Following each solution that corresponds to one product, the catalog rates for recurring and non-recurring charges appear enclosed in brackets. Compound solutions or solutions corresponding to a family of products are followed by ellipses. Sets of solutions that afford two or more capabilities are color-coded for attention, for these may be good candidates to satisfy multiple customer needs. When the solutions cannot be completely displayed in the pane, a horizontal, then a vertical, scroll bar is provided. The SSC can propose solutions appropriate to the circumstances of the customer site from among the possible solutions presented by clicking the solutions. The multi-line help box displays the need trace and solution benefits. Figure 23. shows a typical recommendation window generated by the knowledge-based system.

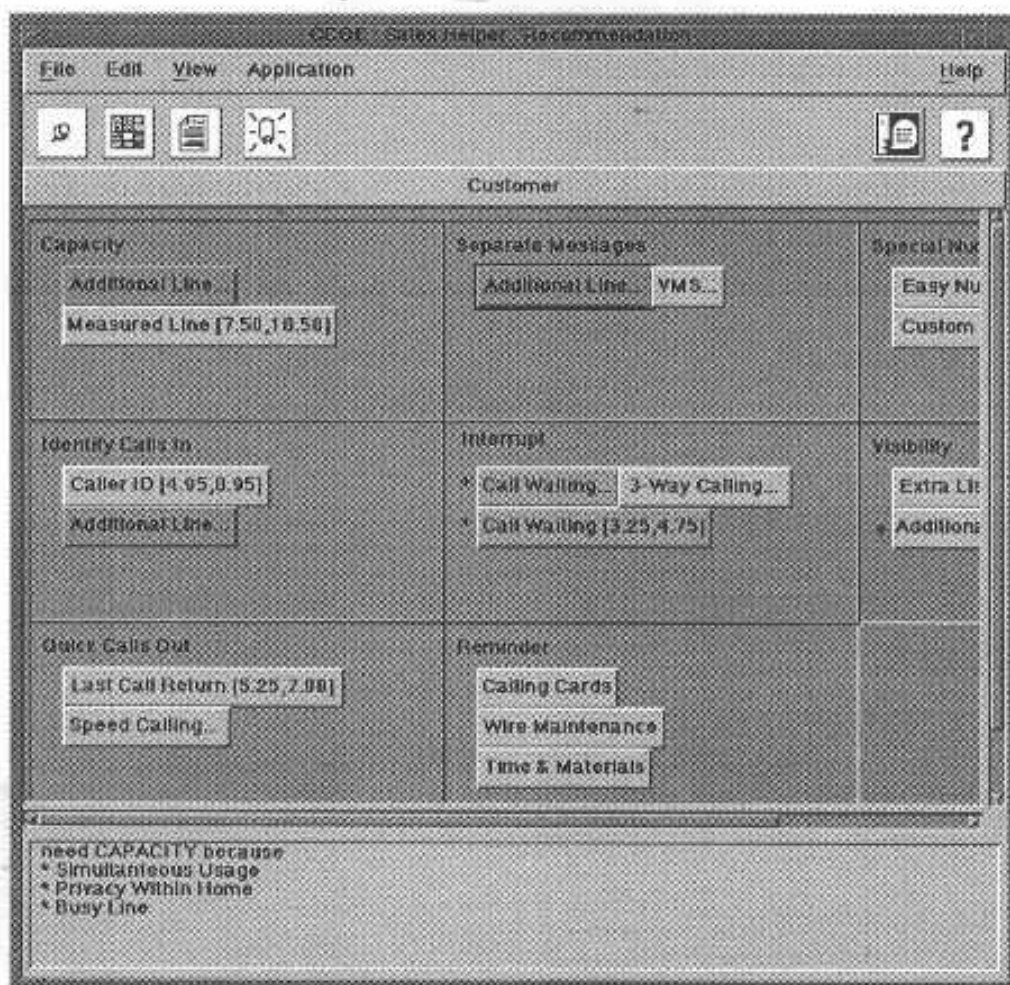


Figure 23. Sales Helper Recommendation Window

This figure includes both one-product and compound solutions. The Additional Line and Call Waiting items illustrate color-coding for solution sets that appear under different capabilities. The plus sign—such as before Call Waiting and Additional Listing—indicates existing products already associated with the customer site. The multi-line help box displays the need trace for Capacity. The SSC controls the display of multi-line help through the Help On/Off icon, then the content of the display through mouse movement.

Selecting a compound solution spawns a secondary dialog box for that solution, as in Figure 24.

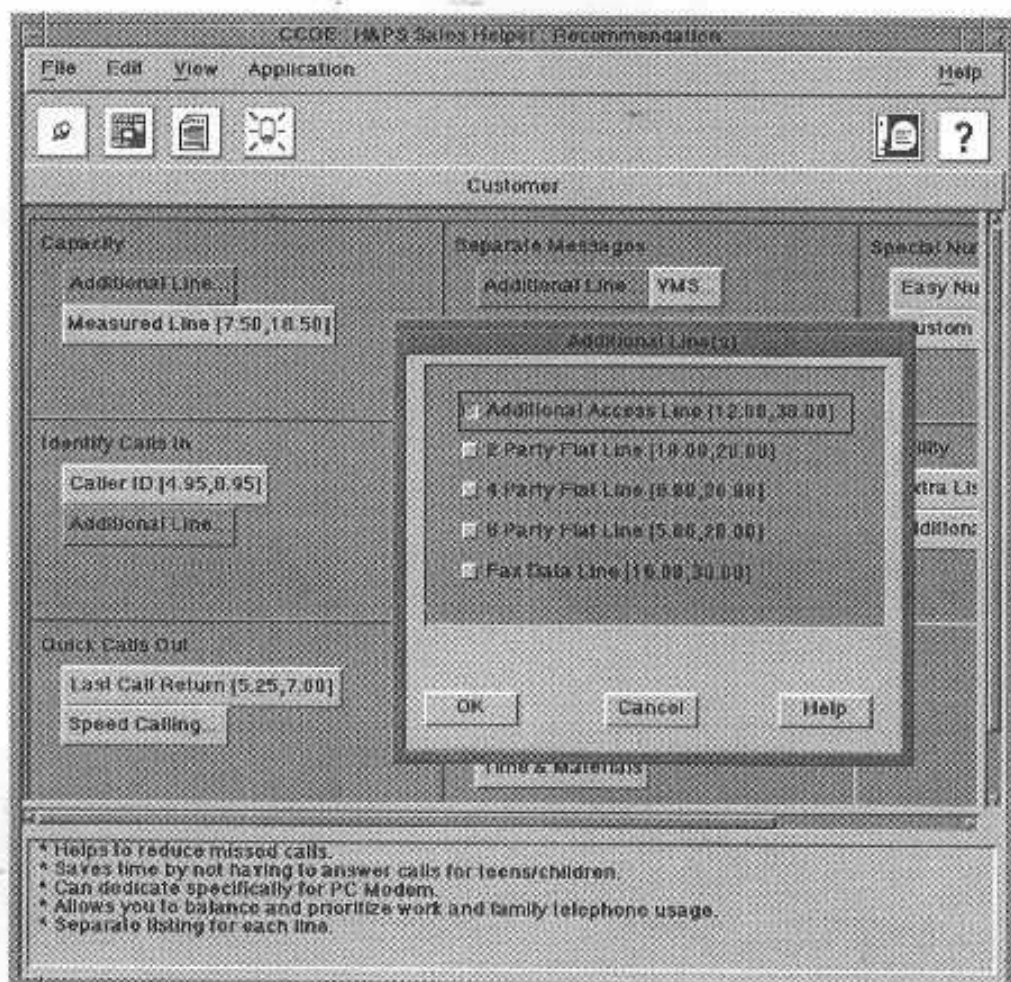


Figure 24. Sales Helper Recommendation Window with Product Choice Dialog Box

In this figure, the dialog box presents the different flavors of the solution Additional Line, and the multi-line help box now displays the benefits of Additional Access Line. This dialog box is titled with the solution name, and is modal; that is, only one solution may be active at a time. It is a link to the Proposed Product list box.

As the SSC chooses one or more products as appropriate to the customer, the choices will be reflected in the Proposed Product list box, illustrated in Figure 25. The SSC can then make use of the proposed product list to navigate the Product Manager.

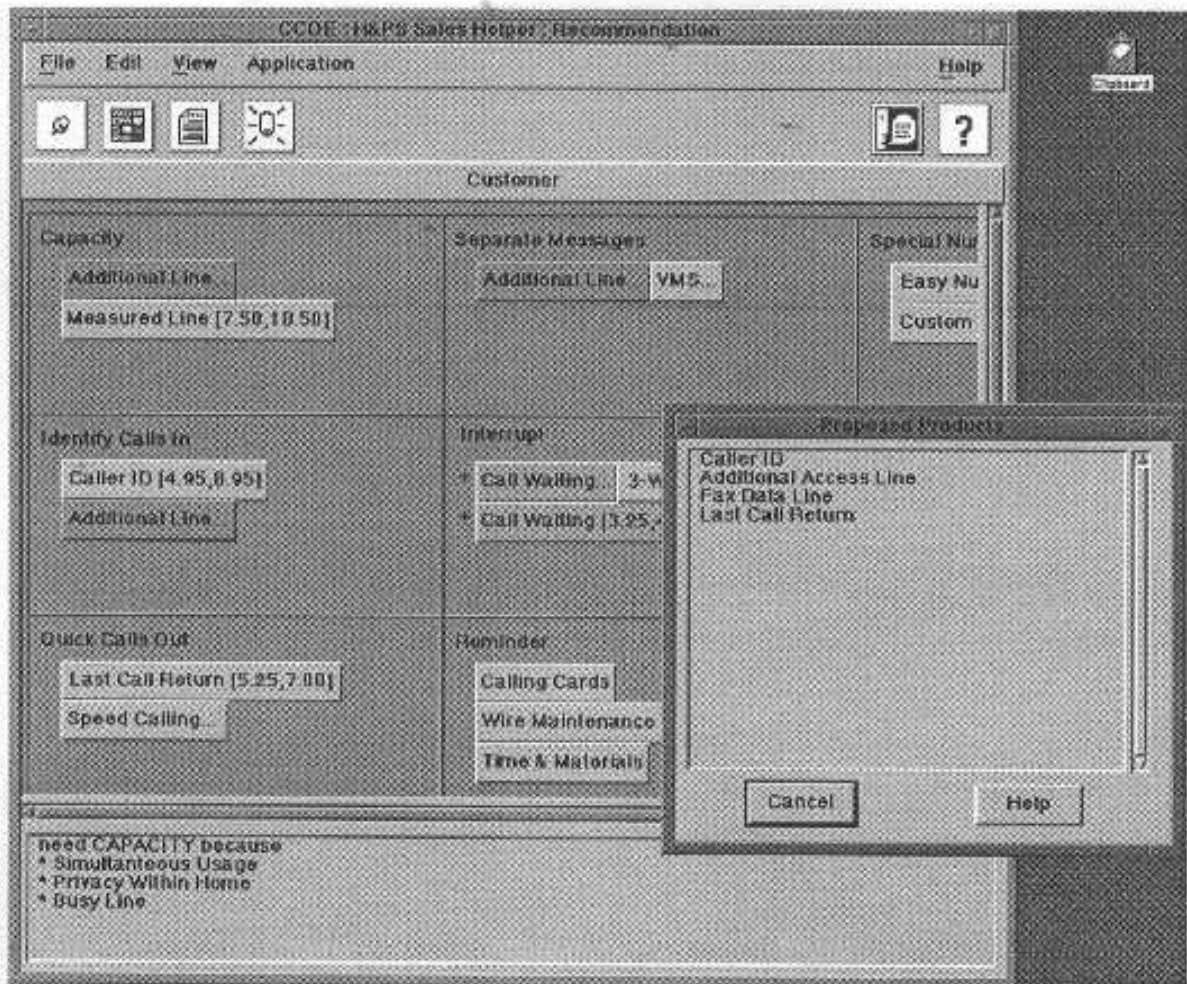


Figure 25. Sales Helper Recommendation Window with Proposed Products Dialog Box

The Proposed Products list box is shown partly superimposed on the Recommendation window. This is a non-modal dialog box that can be moved anywhere on the desktop or sized. The product list is view-only, and the scroll bar becomes active if the list exceeds the limits of the window. The SSC can click on a specific product in the list box to invoke the corresponding Product Manager window. The Proposed Products window remains in the foreground until the SSC selects OK or Cancel to dismiss it.

6.9.1 Recommendation Pane Field Descriptions

#	Field Name	Type	Field Length/ Format	Field Description
1	Capability label(s) (recursive)	text labels	N/A	The capabilities afforded by the recommended solutions according to customer profile information
2	Recommended solutions push button (recursive)	push button	N/A	The recommended solution labels with recurring and non-recurring charges if the mapping of solution to product is one-to-one. The recommended solution labels with ellipses otherwise.

Table 14. Recommendation Window Fields

6.9.2 Recommendation Window Edits

No window edits have been identified.

6.9.3 Recommendation Window CAR Diagram

This section captures interface behavior as specified for each control on the Recommendation window. The following table specifies each control on the Recommendation window, the allowable user actions, and the corresponding system response associated with each of these controls.

For user behavior, the following notation will be used:

click = user clicks on item (button) using mouse button 1.

<Tab> = user tabs to item (button) and then <Return>.

Mnemonic (key) = user activates mnemonic by <Alt> and the key indicated in parentheses.

Accelerator (key) = user activates accelerator by <Control> and the key indicated in parentheses.

Control	User Behavior	System Response
Control Area I: WINDOW MENU BAR		
File cascade button	TBD	TBD
Edit cascade button	TBD	TBD
View cascade button	TBD	TBD
Application cascade button	TBD	TBD
Help cascade button	TBD	TBD
Control Area II: ICONIC PUSH BUTTON BAR		

Table 15. Recommendation Window CAR Diagram

Control	User Behavior	System Response
Profile Page 1 icon	move click	displays field help displays profile page 1
Profile Page 2 icon	move click	displays field help displays profile page 2
Summary icon	move click	displays field help displays profile summary page
Recommendation icon	move click	displays field help Regenerate recommendation from the knowledge-based engine.
Product Manager icon	move click	displays field help Invokes and displays Product Manager
Help Off/On icon	move click	displays field help turns multi-line help area off or on; default is on
Help icon	move click	displays field help TBD
Control Area III: CUSTOMER BANNER		
read only text field (80 characters)	none	displays customer name in all caps
Control Area IV: RECOMMENDATION PANE		
Capability text label(s) (recursive)	move	displays the need trace associated with the capability (in multi-line help area)

Table 15. Recommendation Window CAR Diagram

Control	User Behavior	System Response
Recommended solution button(s) (recursive)	move	displays the benefits of the solution
	click	for one-to-one product mapping, adds the product to the proposed product list. for one-to-many product mapping, spawns a dialog box containing the different flavors of the solution. The SSC can check one or more products from the dialog box. Pressing <OK> adds the chosen product(s) to the proposed product list window. Pressing <Cancel> dismisses the dialog box without adding any products to the proposed list. for compound constituent solutions, presents different push buttons on the same line. Clicking on these push buttons spawns dialog boxes as described in one-to-many mapping.
	double-click (TBD; proposed alternative)	invokes the corresponding Product Manager
Control Area V: MULTI-LINE HELP AREA		
multi-line help	none—system generated	displays field help when the mouse cursor is over the corresponding widget or text label

Table 15. Recommendation Window CAR Diagram

7. Help

Help will be consistent with *OSF/Motif Style Guide Release 1.2*, the *U S West User Interface Guidelines for OSF/Motif*, and conventions given in *Visual Design with OSF/Motif*. The following types of help will be available:

- context-specific help on fields
- help on the objects, operations, or task in the current window
- instructions for using the help system
- help on keyboard mapping
- index of help topics.

The user will access help from a window's menu bar, from a button in a dialog box, and from the keyboard. Any window that does not have a menu bar will access help through a help command button.

7.1 Context-Sensitive Helps

Context-sensitive helps tell users about the functions available at the point in the interface where a user has requested help. The context-sensitive help window is a standard Motif main window. It is draggable, resizable, closeable, and has a text region. If the help information fits within the maximum help window size, there are no vertical and horizontal scroll bars. If the help information doesn't fit within the maximum help window size, active vertical and horizontal scroll bars display.

Helps developed with HyperHelp, Bristol Technology's Help System Development Kit for graphical UNIX environments (Bristol Technology Inc.) will provide online context-sensitive hypertext help. HyperHelp provides a full-featured Windows-style help system that is compatible with any Motif or OpenLook application. A HyperHelp program displays online help information about the application that is currently active in a separate, moveable window.

The process of creating a custom HyperHelp system for an application includes two basic processes. The first concentrates on the help content, and the second on integrating the help system into the application.

7.1.1 Developing Help Content

Phases of help development with HyperHelp include:

1. **identifying what helps are needed in the application**

Task analysis, usability testing, and user input provide the information to determine what helps the SSCs need and to establish a hierarchy of help information based on frequency of use and criticality. This information assists in determining what helps to provide and how and where in the application to provide helps.

2. **writing the help text information**

The next phase of help development involves composing the text information for the help messages. The context in which help is sought determines the content of the messages. The syntax, style, and language level must be tailored to the users' needs. Determining and writing content should be performed by Human Factors and supported by technical writers.

3. **coding and compiling the help text information into an executable HyperHelp program**

Once it has been determined what helps are needed and how critical these helps are, decisions can be made about where and how to place help information in the help hyperspace. HyperHelp's hypertext can provide help for SSCs at all levels of expertise and experience. The CCOE developers will code and compile the help text information into an executable HyperHelp program.

The Human Factors team members must identify:

- the needed help topics
- the presentation of each topic
- the syntax and language level of the text
- the organization of the information hierarchy—the network of information and connective nodes.

This work should be carried out in partnership with the technical writing staff.

7.1.2 Integrating the HyperHelp Program

CCOE links HyperHelp to the Customer Manager window and the other windows and dialog boxes through Help push buttons and Help menu options. The CCOE team programmers must provide the software link between the parent CCOE application(s) and the help system developed in HyperHelp.

The CCOE programmers and writers must develop a unique list of identifiers for:

- each menu option
- each graphic tool, such as icons
- each dialog or other feature associated with a special context-sensitive help topic.

The CCOE programmers must insert the function that, when called, tells the parent application to run the HyperHelp Viewer application and specifies which help topic to display.

7.2 Generic Helps

HyperHelp Viewer, a separate application that provides help for the main application in a separate window, also provides generic helps. Generic helps describe the features and characteristics of the user interface. Users can access generic helps from the options on the parent application's Help menu or by selecting an object with the mouse and pressing <Shift> + <F1>. The HyperHelp Viewer window opens, displaying the appropriate context-specific help topic.

Development of generic helps will follow the process described above for the development of helps.

7.3 Tutorials

Mass markets have responsibility for selecting a vendor to develop computer-based training. At this time it does not appear that the training will be integrated with the application. It will be separately accessible from the desktop menu.

8. Error Conditions

8.1 Error Handling Philosophy

CCOE reduces the likelihood of errors by autopopulating data entry fields, minimizing the amount of typing required, and allowing the user to select only valid items. CCOE performs field-related error checking, so the user making an error gets immediate feedback. The required fields are visually distinct, reminding the user to complete them. Because the user can move freely between and within windows without completing all fields, the check for cross-related fields is done upon user request.

A pop-up Error dialog box displays Error messages. Error dialog boxes stop user interaction with the application until they are dismissed. Error dialog boxes correspond to *OSF/Motif Style Guide Revision 1.2*, the *U S WEST User Interface Overview for OSF/Motif*, and conventions given in *Visual Design with OSF/Motif*.

Some errors detected by CCOE may be reported asynchronously, that is, not at the time the user makes the error or requests an error check. These messages do not interrupt the user's work. CCOE informs the user in a noticeable manner that errors have been found, but that notice does not obstruct the current work. The user must issue a command to display the Error Manager window (Figure 26.) to see these error message(s).

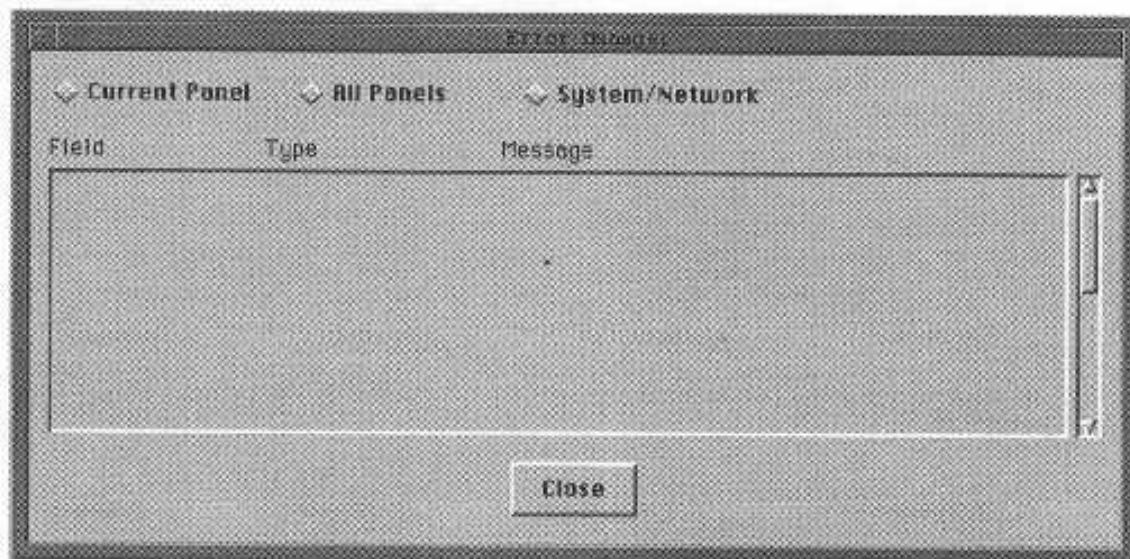


Figure 26. Error manager

The behavior of the Error Manager has not been determined at this time. Behavior for this part of CCOE must await certain architectural decisions before proceeding.

8.2 Error Messages

TBD

9. Usability Criteria

For the purposes of this document, a usability problem is globally defined as anything in the user interface that prevents the SSC from completing a task with reasonable effort and within a reasonable time. The initial usability testing will focus on eliciting users' subjective evaluation of the CCOE user interface and will draw on their ability to identify usability problems.

The primary goal of this evaluation is to determine, from a user's perspective, if CCOE provides a clear, concise, and efficient user interface that delivers the functionality required by the task. When focusing on the details of an application GUI, it is important not to overlook the larger issues that ultimately determine whether a system is likely to be successful. For example, fixing all of the small usability problems may still yield an unusable system if it lacks an overall conceptual model that the end-user finds useful and productive.

Initial testing will be formative evaluation. The main objective of formative evaluation is to identify which components of the user interface support task performance, which hinder task performance, and how the user interface can be improved. Formative evaluation to help improve the user interface is part of the iterative design process.

To ensure the quality of the usability data collected, several methods and measures have been selected from a standard set of procedures for usability evaluations. The selected methods and measure are:

- errors or dialogue failures
- verbal protocols
- attitude measures
- conceptual walkthroughs
- focus groups
- time measurements
- interviews.

These methods were selected because they do not require a full functional prototype to produce useful data. General descriptions of each method and the procedures involved follow.

9.1 Errors or Dialog Failures

Errors or dialog failures that occur at the user interface are one of the most useful sources of information in an evaluation. These errors reveal where problems exist within a system's user interface. Analysis of errors or dialog failures may also suggest the cause of the problem. In essence, by studying why the system and the user failed to communicate properly a solution to the problem may appear.

Errors or dialogue failures are often associated with areas in the user interface that do not match the user's needs or the user's understanding of the task. In order to diagnose problems, errors or dialog failures can be classified into different types. An error in the user's intention is a mistake, while an error in carrying out an intention is a slip. Mode errors occur when the user performs an action that is inappropriate for the current mode. Finally, a description error occurs where there is insufficient specification for an action, and the confusion leads to the wrong action.

9.2 Verbal Protocols

Verbal protocols are records of what a user said either during or after performing a task. There are two types of verbal protocols: concurrent or "think aloud," and retrospective. A think aloud protocol is one that is taken while the user is performing a task. When this procedure is utilized, users are asked in advance to verbalize their thoughts and intentions (to "think aloud") as they interact with an application or device. A retrospective protocol is taken once the task has been completed. Only concurrent or "think aloud" protocols will be used in evaluating the CCOE interface.

9.3 Attitude Measures

Questionnaires will solicit the users' perceptions of such application characteristics as learnability, ease of use, acceptability, and functionality. The questionnaires will include open-ended questions about the system. Open-ended questions are designed to give the users an opportunity to pinpoint those aspects of the system they felt were especially helpful, as well as those parts of the system they found confusing.

9.4 Conceptual Walk-through

A conceptual walk-through provides a means for users to give feedback to the design team and to identify areas in the user interface they feel are problematic or difficult to understand. This process enhances the validity of the design and increases end user acceptance. In a conceptual walk-through, participants and a coach examine each screen of the user interface

for its overall presentation. A conceptual walk-through lets users describe and explain in detail those aspects of the user interface they like and those aspects that they find difficult to understand.

9.5 Focus Groups

Focus groups provide a structured forum in which users can express their attitudes and concerns regarding various aspects of the system. It is an excellent opportunity to gather information from a group. This group process often generates different ideas, feedback, and perspectives than a one-on-one interaction between the designer and an end-user.

The focus group method is often used in combination with individual feedback sessions. The focus group methodology is especially useful for several reasons:

- Group synergy provides a different perspective on the usability of the application.
- People feel less inhibited about reporting negative information in group settings.
- Designers or evaluators are less able to influence responses.

The possibility that a designer may unintentionally elicit only positive feedback from users in individual sessions always exists.

It is important to consider the composition of the group. Sometimes combining members of unequal rank can inhibit some members of the group from expressing their true opinions. Yet having a homogeneous group of users may not reveal problems that other—either less or more experienced—users might have.

9.6 Time and Usage Measurements

Time measurements provide an empirical criteria of relating users' tasks with time required to complete the tasks. The time spent in a window (or a group of windows as related to specific tasks) can be measured either by the program or through observation with stop watches.

Usage measurements determine the frequency of the various optional components of the software used by the SSCs. Both methods can be used at different stages of system development. A baseline can be used as a reference for comparing measurements conducted at different points in time.

9.7 Usability Test Plan

In the usability studies on CASH in Phoenix and Omaha, we conducted time and usage measurements across different types of calls. A focus group was conducted in Omaha. The SSCs in both sites participated in usability surveys and informal interviews.

SSCs from the Denver Center of Customer Service also participated in a study of the CCOE Sales Helper (H&PS) Prototype in July, 1994.

More usability assessments will be conducted as time, resource, and budget permit.

9.7.1 Time Frame

The usability study for Home Office Channel (SBG) in Phoenix was conducted in June, 1994 and Market Place Channel (H&PS) in October, 1994. Reports have been issued for both studies.

9.7.2 Resources

For CCOE Sales Helper Prototype usability studies, resources include a portable computer with all necessary software to run the prototype and ten to fifteen service representatives per site. Participating SSCs will spend a total of 1 1/2 hours each for the prototype evaluation. An additional one hour is needed from all SSCs participating in the prototype evaluation for the focus group evaluation.

In the field studies of CASH for Home Office and Market Place Channel, stop watches will be needed to time the calls.

9.7.3 Sites

Future sites of assessment will be determined.

9.7.4 Evaluation Procedure

- briefly introduce CCOE and purpose of walkthrough = 10 minutes.
- gather verbal reports from SSCs during their interaction with the prototype ≈ one hour.
- gather reports concerning likes and dislikes for individual screens ≈ 30 minutes.
- administer a brief questionnaire to SSCs ≈ 10 minutes.
- debrief SSCs at the end of the evaluation ≈ 10 minutes.
- conduct one-hour focus groups with SSCs providing feedback on the prototype.

9.7.4.1 Introduction

During the introduction, the coach introduces him/her self to the participants and explains the purpose of the evaluation. The coach must:

- clearly state that the evaluation is targeted at the application, both its good and bad aspects, and not at the SSC's ability to use the prototype
- inform SSCs that the test results will be used to improve the user interface and that comments related to all aspects of the application, no matter how small, are useful
- inform SSCs that all information they provide is confidential
- answer all questions SSCs have before starting the evaluation
- remind SSCs to ask questions during the evaluation
- instruct SSCs that their participation is voluntary and that they may quit at any time during the evaluation.

9.7.4.2 Verbal Reports

While the SSCs interact with the prototype, they will "think aloud" to provide verbal feedback to the coach. The coach addresses any questions and records all relevant comments made by the SSCs.

9.7.4.3 Screen Feedback Questionnaire

The screen feedback booklet contains a hard copy of each screen in the prototype, followed by a page of three closed-ended questions that assess likability, usability, and functional completeness. The SSCs answer these three questions using a seven-point Likert scale (1=strongly disagree to 7=strongly agree). An open-ended question, asking for general comments on each screen, follows.

9.7.4.4 Attitude Questionnaire

The attitude questionnaire is a 10-item scale designed to assess general usability of CCOE. The SSCs answer using a seven-point Likert scale (1=strongly disagree to 7=strongly agree).

9.7.4.5 Debriefing

During the debriefing, the coach asks SSCs for general feedback on the application and for any additional suggestions. The coach must inform the SSCs that not all suggestions result in user interface design changes; that often different users make contradictory suggestions; and that all suggestions must be accounted for before specific design changes are made.

9.7.4.6 Focus Group

The coach conducts the focus groups, providing a structured forum in which users can express their attitudes and concerns regarding the system. A standard structure will be used across all focus groups, with an agenda of what should be covered during the group session. An observer, not the coach, will record all comments and responses.

10. Performance and Quality Criteria

TBD based on approval and sign-off for *Customer Contact Operating Environment (CCOE Mass Markets) User Requirements* document.

11. Field Evaluation Plan

This section describes the criteria and procedures used to evaluate the performance of the system.

Usability assessments on CASH have been conducted in the Home Office Channel in Phoenix and the Market Place Channel in Omaha. Results were reported elsewhere.

11.1 Method of Evaluation

TBD

11.2 Participating Organizations

TBD

12. Issues

- Whether the Sales Helper Icon is available from the Customer Product Viewer only, or whether it is included in more screens (or as many as possible screens) in CCOE.
- All the pull-down menus of the screens have to be determined.
- # of Jacks in Profile Information may not be legal to ask.
- Link between Recommendation and Product Editor is still tenuous.
- Recommendation window may have too much information for the SSC to handle.
- Whether invoking Sales Helper from an unknown site is beneficial.
- Non-standard use of double-click of pushbuttons to access Product Manager:
The SSC can move from the Recommendation window to the ordering module by double-clicking on any pushbutton representing a solution or individual component solution, if it is a compound solution. The action will invoke the corresponding Product Manager screen. If the solution has different flavors, the Product Manager will show the first screen associated with that generic solution. If necessary, the SSCs may scroll to different pages to locate a specific product in the Product Manager. In this expert mode of activating Product Manager, choosing a solution will not expand it into products or put it in the Proposed Products List.
- Whether Profile Summary pane can allow updating the known profile items, as illustrated in Figure 27.

CCOE - NAPS Sales Helper - Summary

File Edit View Application Help

Customer

Service Status		Phone Users		Customer Needs	
	Yes No		Yes No		Yes No
Moving	☒ ☐	# Users	4	Can't Miss Calls	☒ ☐
Layout		Spouse	☒ ☐	Home Members Unique ID	☒ ☐
Home Type	Multi-Level	# Children	2		
# Jacks	5	Child under 5 Yrs	☒ ☐		
Equipment		Child 11 - 18 Yrs	☒ ☐		
# Phone Sets	3	Life Style		Yes No	
Modem / Fax	☒ ☐	Commute to Work	☒ ☐		
Usage		Home Office	☒ ☐		
In Local	21+ min day	Concerns			
Toll In LATA	Low	Waiting for Line	☒ ☐		

Last Update
5-25-1995

Profile 1

Profile 2

Figure 27. Proposed Summary Window with Update Capability

Appendix A. H&PS Category and Item Definitions

Table 16. contains the H&PS profile category and item definitions.

Category	Profile Item	Definition	Widget
Service Status		Determine the type of service change requested.	text label
	New Service	Customer's initial order for new service with no working service at this location	text label custom tri-state toggle buttons (2)
	Moving	Customer moving from one location to another and wants to transfer service.	text label custom tri-state toggle buttons (2)
	Total Disconnect	Customer no longer needs service at this location.	option menu -Move fr Terr -Move fr State -Furnished Svc -Cost -Deceased -Unknown
Layout		Determine the layout of the home and the phone service there.	text label
	Home Type	Structure of home including levels, living space, and work areas	option menu -One Level -Multi Level -Apartment -Trailer -Other -Unknown
	Detached Locs	A location such as garage, workshop, carriage house etc.	text label custom tri-state toggle buttons (2)

Table 16. Profile Categories and Items (H&PS)

Category	Profile Item	Definition	Widget
	# of Jacks	Number of existing jacks in a household	option menu -0 -1 -2 -3 -4 -5+ -Unknown
	Alarm	A security system or alarm is present at customer location.	text label custom tri-state toggle buttons (2)
Call Motivated by		Determine whether the call is motivated by advertisement or promotion	text label
	Media	What advertising medium prompted the customer to call U S WEST?	option menu -TV -Radio -Direct Mail -Newspaper -Insert -Word of Mouth -Other -Unknown
	Promotion	Which product promotion prompted the customer to call U S WEST?	option menu -A -B -C -Unknown
Equipment		Determine what equipment the customer has at home.	text label
	# of Phone Sets	Number of phone sets a customer plans to use on a line	option menu -1 -2 -3 -4+ -Unknown
	Ans Mch/Svc	Customer has either an answering machine or service	text label custom tri-state toggle buttons (2)
	Pager	Customer has a signalling device	text label custom tri-state toggle buttons (2)

Table 16. Profile Categories and Items (H&PS)

Category	Profile Item	Definition	Widget
	Cellular / Mobile	Customer has cellular or mobile service	text label custom tri-state toggle buttons (2)
	Modem / Fax	Customer has modem and/or fax machine	text label custom tri-state toggle buttons (2)
Usage		Determine the types of calls and the approximate length of calls.	text label
	In Local	The amount of time the customer spends on incoming local calls	option menu -0-10 min day -11-20 min day -21+min day -Unknown
	Out Local	The amount of time the customer spends on outgoing local calls	option menu -0-10 min day -11-20 min day -21+min day -Unknown
	Toll in LATA	To what extent the service is used for long distance calls within the LATA (area code)	option menu -High -Medium -Low -Unknown
Phone Users		Determine who uses the phone and the budget for phone services.	text label
	# of Users	Total number of users using the service within a household	option menu -1 -2 -3 -4+ -Unknown
	Spouse	Legal spouse	text label custom tri-state toggle buttons (2)
	Roommates	Adults sharing living quarters	text label custom tri-state toggle buttons (2)

Table 16. Profile Categories and Items (H&PS)

Category	Profile Item	Definition	Widget
	# of Children	Indicates the number of children under the age of 18	option menu -0 -1-2 -3-4 -5+ -Unknown
	Child Under 5 Yrs	Any children under 5 years of age	text label custom tri-state toggle buttons (2)
	Child 5 - 10 Yrs	Any children ages 5 - 10 years	text label custom tri-state toggle buttons (2)
	Child 11 - 18 Yrs	Any children ages 11 - 18 years	text label custom tri-state toggle buttons (2)
	Family in Other Locs	Family such as college children, or step-children living with other parents	text label custom tri-state toggle buttons (2)
	Additional Users	Additional household users such as parents, extended family, etc.	option menu -Parents -Extended Family -Others -Combination -Unknown
	Special Needs	Extenuating circumstances or physical challenges, requiring special solutions	text label custom tri-state toggle buttons (2)
	Budget	Amount of money (revenue) allocated monthly towards U S WEST services	option menu -<\$20 mo -\$21-50 mo -\$51-75 mo -75+ mo -Unknown
Life Style		Determine the life style factors.	text label

Table 16. Profile Categories and Items (H&PS)

Category	Profile Item	Definition	Widget
	Commute to Work	One or more household members are employed at a location other than home.	text label custom tri-state toggle buttons (2)
	Home Office	One or more household members conduct business from home.	text label custom tri-state toggle buttons (2)
	Travels Frequently	The customer travels for either business or personal reasons.	text label custom tri-state toggle buttons (2)
	Out and About	Customer is away from home frequently, running errands etc.	text label custom tri-state toggle buttons (2)
	Extended Absence	Customer is gone for a temporary period of time.	text label custom tri-state toggle buttons (2)
Concerns		Determine concerns the customer has regarding their phone service.	text label
	Busy Line	The line is occupied with little opportunity to receive calls.	text label custom tri-state toggle buttons (2)
	Waiting for Line	Customer spends too much time waiting to use the line.	text label custom tri-state toggle buttons (2)
	Annoyance Calls	Repeated calls are irritating, threatening or obscene.	text label custom tri-state toggle buttons (2)
	Lost Messages	Customer has not received, erased, or lost messages.	text label custom tri-state toggle buttons (2)
	Phone Tag	Leaving messages but unable to communicate verbally	text label custom tri-state toggle buttons (2)

Table 16. Profile Categories and Items (H&PS)

Category	Profile Item	Definition	Widget
	Missed Calls	Calls are missed when away from home.	text label custom tri-state toggle buttons (2)
	Misdialed #s Out	Reaching incorrect numbers when calling out	text label custom tri-state toggle buttons (2)
Customer Needs		Determine the telecommunication needs of the customer.	text label
	Easy to Reach	Customer needs to be available to calling parties.	text label custom tri-state toggle buttons (2)
	Can't Miss Calls	Important to receive all calls	text label custom tri-state toggle buttons (2)
	Privacy Within Home	Customer wants to maintain confidentiality within household.	text label custom tri-state toggle buttons (2)
	Home Members Unique ID	Customer wants to maintain individual identity within household.	text label custom tri-state toggle buttons (2)
	Save Messages	Customer wants the ability to keep messages for present or future access.	text label custom tri-state toggle buttons (2)
	Screen Calls	Customer wants the ability to choose certain calls or callers.	text label custom tri-state toggle buttons (2)
	Simultaneous Usage	More than one person/machine, uses/needs to use the line at the same time.	text label custom tri-state toggle buttons (2)

Table 16. Profile Categories and Items (H&PS)

Category	Profile Item	Definition	Widget
	Restrict In Local	Restrict or limit incoming local calls	text label custom tri-state toggle buttons (2)
	Restrict In Toll	Restrict or limit incoming long distance calls	text label custom tri-state toggle buttons (2)
	Restrict Out Toll	Restrict or limit outgoing long distance calls	text label custom tri-state toggle buttons (2)
	Conference	Customer needs the ability to talk to more than one person at the same time.	text label custom tri-state toggle buttons (2)
	Frequently Dialed #s	The same numbers are dialed often.	text label custom tri-state toggle buttons (2)
	Easy Callback	Customer needs a quick and easy method of returning calls.	text label custom tri-state toggle buttons (2)
	Easy Number	Customer needs easy telephone numbers to recognize by both calling parties and self.	text label custom tri-state toggle buttons (2)

Table 16. Profile Categories and Items (H&PS)

Appendix B. SBG Category and Item Definitions

Table 17. contains the SBG profile category and item definitions.

Category	Profile Item	Definition	Widget
Business Info		Determine the business information of customer at the site chosen.	text label
	New / Startup	New or startup business (at a new site) or business starting a new site	text label custom tri-state toggle buttons (2)
	Moving	Business moving from one location to another now	text label custom tri-state toggle buttons (2)
	Growth	Whether business is growing, maintaining or downsizing	option menu -Expand -Maintain -Downsize
	Multi-Users	Different people using phone lines and facilities at the customer site	text label custom tri-state toggle buttons (2)
	Multi-Use Line	Mixed usage with the equipment of the business	text label custom tri-state toggle buttons (2)
	Work Home Occasionally	Employees of business work at home occasionally.	text label custom tri-state toggle buttons (2)
Call Motivated by		Determine whether the call is motivated by advertisement or promotion.	text label

Table 17. Profile Categories and Items (SBG)

Category	Profile Item	Definition	Widget
	Media	What advertising media prompted the customer to call U S WEST?	option menu -TV -Radio -Direct Mail -Newspaper -Insert -Word of Mouth -Other -Unknown
	Promotion	Which product promotion prompted the customer to call U S WEST?	option menu -A -B -C -Unknown
Layout		Determine the layout of the customer site.	text label
	# of Staff	Number of employees at the customer site	option menu
	Depts/Offices	Multiple departments or offices within the customer site	text label custom tri-state toggle buttons (2)
	Work Area	Multiple work areas within the customer site	text label custom tri-state toggle buttons (2)
	Multi-Premises	Business has more than one site or location.	text label custom tri-state toggle buttons (2)
Concerns		Determine concerns the customer has regarding their phone service.	text label
	Busy Line	The line is occupied with little opportunity to receive calls.	text label custom tri-state toggle buttons (2)
	Waiting for Line	Customer spends too much time waiting to use the line.	text label custom tri-state toggle buttons (2)

Table 17. Profile Categories and Items (SBG)

Category	Profile Item	Definition	Widget
	Annoyance Calls	Repeated calls that are irritating, prank, harassing, threatening or obscene	text label custom tri-state toggle buttons (2)
	Lost Messages	Customer has not received, erased, or lost messages.	text label custom tri-state toggle buttons (2)
	Phone Tag	Leaving messages but unable to communicate verbally	text label custom tri-state toggle buttons (2)
	Missed Calls	Missed telephone calls	text label custom tri-state toggle buttons (2)
	Priority Callers	Business has priority or special clients or suppliers.	text label custom tri-state toggle buttons (2)
	Privacy	Business needs private or separate means of communication or storage of messages	text label custom tri-state toggle buttons (2)
	Annoyance Calls	Business receives obscene, harassing or prank calls	text label custom tri-state toggle buttons (2)
Easy Calling		Business needs easy calling features in and out	text label
	Easy to Reach	Clients can easily reach business potentially at no cost to them.	text label custom tri-state toggle buttons (2)
	Easy Number	Clients of the business want to easily find, locate and remember the business.	text label custom tri-state toggle buttons (2)
	Save Messages	Staff of the business want to be able to save incoming messages.	text label custom tri-state toggle buttons (2)

Table 17. Profile Categories and Items (SBG)

Category	Profile Item	Definition	Widget
	Freq Dialed #s	Staff of the business frequently dial the same number.	text label custom tri-state toggle buttons (2)
	Conference	Staff of the business talk with more than one person at a time on phone from time to time.	text label custom tri-state toggle buttons (2)
	Info Only	Business provides its clients with informational messages on the phone.	text label custom tri-state toggle buttons (2)
	Transfer	Business has a need to transfer calls.	text label custom tri-state toggle buttons (2)
Identify Calls		Business needs to identify incoming or outgoing calls.	text label
	Screen	Staff of business can selectively answer calls.	text label custom tri-state toggle buttons (2)
	Track Calls In	Business can track incoming calls or measure market and advertising.	text label custom tri-state toggle buttons (2)
Control Calls		Business needs to control incoming or outgoing calls.	text label
	Restrict In	Business needs to control inbound and bill-to or third party calls.	text label custom tri-state toggle buttons (2)
	Restrict Out	Business needs to control outbound calls.	text label custom tri-state toggle buttons (2)
	Urgent	Business needs to respond to calls immediately.	text label custom tri-state toggle buttons (2)

Table 17. Profile Categories and Items (SBG)

Category	Profile Item	Definition	Widget
Ans Methods		Determine the method used by the business to answer the phone.	text label
	Self	The caller will answer or answers the phone most of the time.	text label custom tri-state toggle buttons (2)
	Others	Other people (staff) will answer or answer the phone most of the time.	text label custom tri-state toggle buttons (2)
	After Hours	Business will answer phone before or after normal business hours.	text label custom tri-state toggle buttons (2)
Out of Office		Business has out-of-office communication needs.	text label
	Work out of Office	Business has employees working out of office.	text label custom tri-state toggle buttons (2)
	Voice	Employees of business need to communicate while out of office.	text label custom tri-state toggle buttons (2)
	Message Notify	Employees of business need message notification when out of office.	text label custom tri-state toggle buttons (2)
	On Call	Employees of business need to respond to some requests promptly.	text label custom tri-state toggle buttons (2)
In/Out Usage		Determine the in and out phone usage of the business	text label
	Local In	Local usage in	option menu -High -Medium -Low -Unknown

Table 17. Profile Categories and Items (SBG)

Category	Profile Item	Definition	Widget
	Local Out	Local usage out	option menu -High -Medium -Low -Unknown
	IntraLATA In	IntraLATA usage in	option menu -High -Medium -Low -Unknown
	IntraLATA Out	IntraLATA usage out	option menu -High -Medium -Low -Unknown
	InterLATA In	InterLATA usage in	option menu -High -Medium -Low -Unknown
	InterLATA Out	InterLATA usage out	option menu -High -Medium -Low -Unknown
	Data In	Data usage in (fax, modem, credit card machine)	option menu -High -Medium -Low -Unknown
	Data Out	Data usage out (fax, modem, credit card machine)	option menu -High -Medium -Low -Unknown
Usage		Pattern of line usage of business	text label
	Seasonal	Business has seasonal peak usage (month or season).	text label custom tri-state toggle buttons (2)

Table 17. Profile Categories and Items (SBG)

Category	Profile Item	Definition	Widget
	Extended Absence	Business closes temporarily during the day for delivery or miscellaneous chores.	text label custom tri-state toggle buttons (2)
Equipment		Determine the current and future equipment of the business.	text label
	# Phone Sets	Number of phone sets the business has	option menu -0-1 -2-4 -5-7 -8+ -Unknown
	Cordless	Business owns, buys, plans to buy or has problems with cordless sets.	option menu -Own -Plan -Problem -None -Unknown
	Cellular	Business owns, buys, plans to buy or has problems with cellular or mobile services.	option menu Own -Plan -Problem -None -Unknown
	Pager	Business owns, buys, plans to buy or has problems with pager.	option menu Own -Plan -Problem -None -Unknown
	Fax	Business owns, buys, plans to buy or has problems with fax machines.	option menu Own -Plan -Problem -None -Unknown
	Modem	Business owns, buys, plans to buy or has problems with PC modems.	option menu -Own -Plan -Problem -None -Unknown

Table 17. Profile Categories and Items (SBG)

Category	Profile Item	Definition	Widget
	Credit Card Mch	Business owns, buys, plans to buy or has problems with credit card machines.	option menu Own -Plan -Problem -None -Unknown
	Ans Mch	Business owns, buys, plans to buy or has problems with answering machines.	option menu Own -Plan -Problem -None -Unknown
	Ans Svc	The phone is answered by telephone answering service most of the time.	text label custom tri-state toggle buttons (2)
Image		Perception and image the business wants to relay to its market.	text label
	Big	Business wants appearance of large company.	text label custom tri-state toggle buttons (2)
	Customized	Business wants to project a special or personalized image.	text label custom tri-state toggle buttons (2)
Visibility		Business wants visibility in different geographical areas.	text label
	Local	Business wants presence within local calling areas.	text label custom tri-state toggle buttons (2)
	Key Cities	Business wants key cities presence outside their local calling areas.	text label custom tri-state toggle buttons (2)
	State	Business wants state-wide presence.	text label custom tri-state toggle buttons (2)

Table 17. Profile Categories and Items (SBG)

Category	Profile Item	Definition	Widget
	National	Business wants national presence.	text label custom tri-state toggle buttons (2)
Advertising		Determine how the business advertises.	text label
	Phone Book	Business has needs to advertise in phone books.	text label custom tri-state toggle buttons (2)
	Multimedia	Business has needs to advertise in flyers, magazines, newspapers, radio, TV or billboards.	text label custom tri-state toggle buttons (2)
Customer		Determine client base of the business.	text label
	New	Business depends on new clients.	text label custom tri-state toggle buttons (2)
	Repeat	Business depends on repeat clients.	text label custom tri-state toggle buttons (2)
	Walk-ins	Business depends on walk-ins.	text label custom tri-state toggle buttons (2)
	Referrals	Business depends on referrals.	text label custom tri-state toggle buttons (2)

Table 17. Profile Categories and Items (SBG)

Appendix C. Subjective Ease of Use Instrument

Subjective ease of use instruments are currently under evaluation.

Appendix D. Error Messages

TBD

